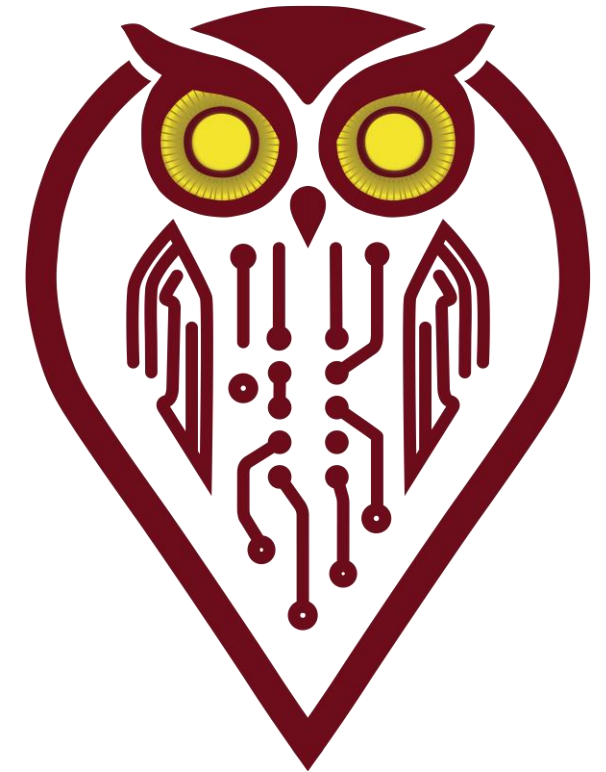


Perception and Intelligence Laboratory

Computer Science Department
Sapienza University of Rome, Italy



Prof. Fabio Galasso
25 September 2026



SAPIENZA
UNIVERSITÀ DI ROMA

PINlab and ItAI scientists



Fabio Galasso

Head of the lab & Full Professor



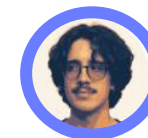
Indro Spinelli

Assistant Professor



Simone Facchiano

Ph.D. Student



Matteo Migliarini

Ph.D. Student



Xavier Del Giudice

Ph.D. Student



Guido Maria D'Amely di Melendugno, Ph.D.

Post-Doc Research Associate



Luca Franco, Ph.D.

Manager at ItAI



Alessio Palma

Ph.D. Student



Luca Romani

Ph.D. Student



Anja Stanić

Ph.D. Student



Matteo Gioia

Ph.D. Student



Leonardo Plini

Ph.D. Student



Xuanhao Fan

Ph.D. Student



Stefano Saravalle

Ph.D. Student



Michele Palma

Ph.D. Student



Valentino Sacco

Ph.D. Student



Aurora Bassani

Ph.D. Student



Pietro Signorino

Ph.D. Student



Paolo Barba

Ph.D. Student



Christian Bianchi

M.Sc. Researcher

PINlab and ItalAI Alumni

- Up to
 - Their own startup
 - A top university, e.g. ETH Zurich, Technical University of Munich
 - Public sector, e.g. Banca D'Italia, Agenzia per la Cybersicurezza nazionale
 - Scaling startups, e.g. WSense, Translated
 - Large tech companies, e.g. Samsung, Panasonic



Alessandro Flaborea, Ph.D.

Co-founder of Procede



Alessio Sampieri, Ph.D.

Senior AI Researcher at Translated



Giovanni Ficarra, Ph.D.



Luca Collorone, Ph.D.

Post-Doc Research Associate at
Technical University of Munich



Luca Scofano, Ph.D.

Senior AI Researcher at Translated



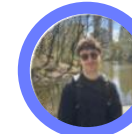
**Muhammad Rameez Ur
Rahman, Ph.D.**

Post-Doc Research Associate at Ca'
Foscari University of Venice



Stefano D'Arrigo, Ph.D.

AI Researcher @ DeepPlants



Edoardo De Matteis, Ph.D.

AI Expert at Agenzia per la
Cybersicurezza Nazionale



Laura Laurenti, Ph.D.

Post-Doc Research Associate at ETH
Zurich



Paolo Mandica, Ph.D.

AI Research Scientist at Samsung



Massimiliano Pappa, Ph.D.



Daniele Trappolini, Ph.D.





Briefly about myself

- Career
 - ▶ Ph.D. and Pdoc in Cambridge
 - ▶ Research associate at MPI
 - ▶ Head of Computer Vision Dept at OSRAM
 - ▶ Full Professor at **Sapienza**
 - ▶ Co-founder and CTO of **ItalAI** S.r.l.
- Perception and Intelligence Lab (PINLab) at **Sapienza**
 - ▶ Computer Vision and Machine Learning
 - ▶ Generative AI and Robotics
 - ▶ Uncertainty, Interpretable and Safe AI
- **ItalAI** Sapienza start-up
 - ▶ Technological innovation in real-world scenarios





ST JOHN'S COLLEGE
UNIVERSITY OF CAMBRIDGE



UNIVERSITY OF
CAMBRIDGE

- 2005-2009
PhD in Engineering
- 2009-2011
PDoc Researcher

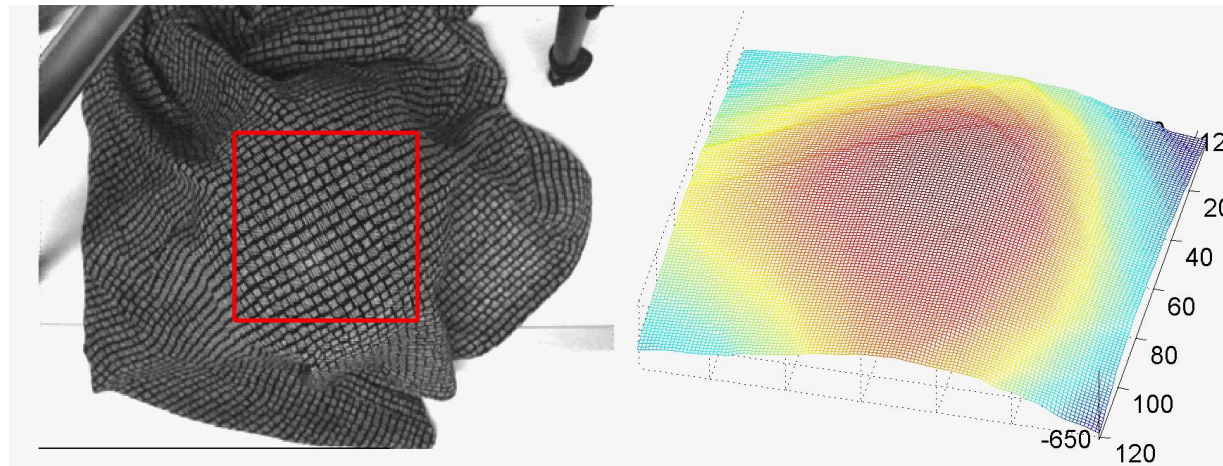
Textures and 3D reconstruction

BMVC'07, ISVC'07, ISVC'08, CVPR'09

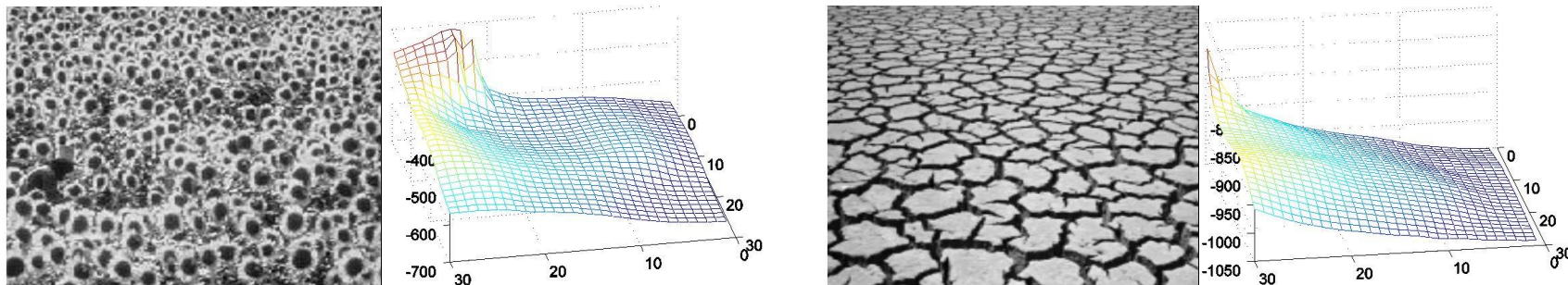


UNIVERSITY OF
CAMBRIDGE

- Leverage texture to recover the shape
 - In controlled environments



- In the “wild”



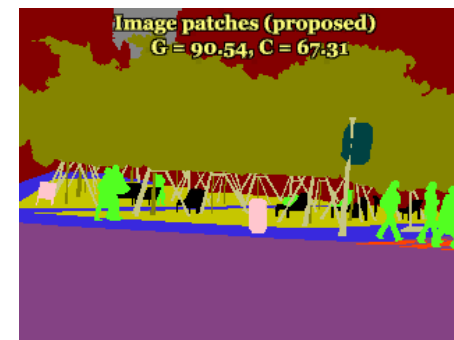
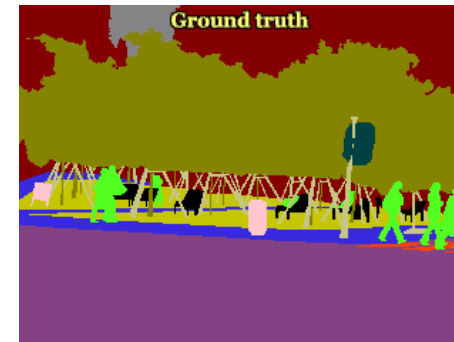
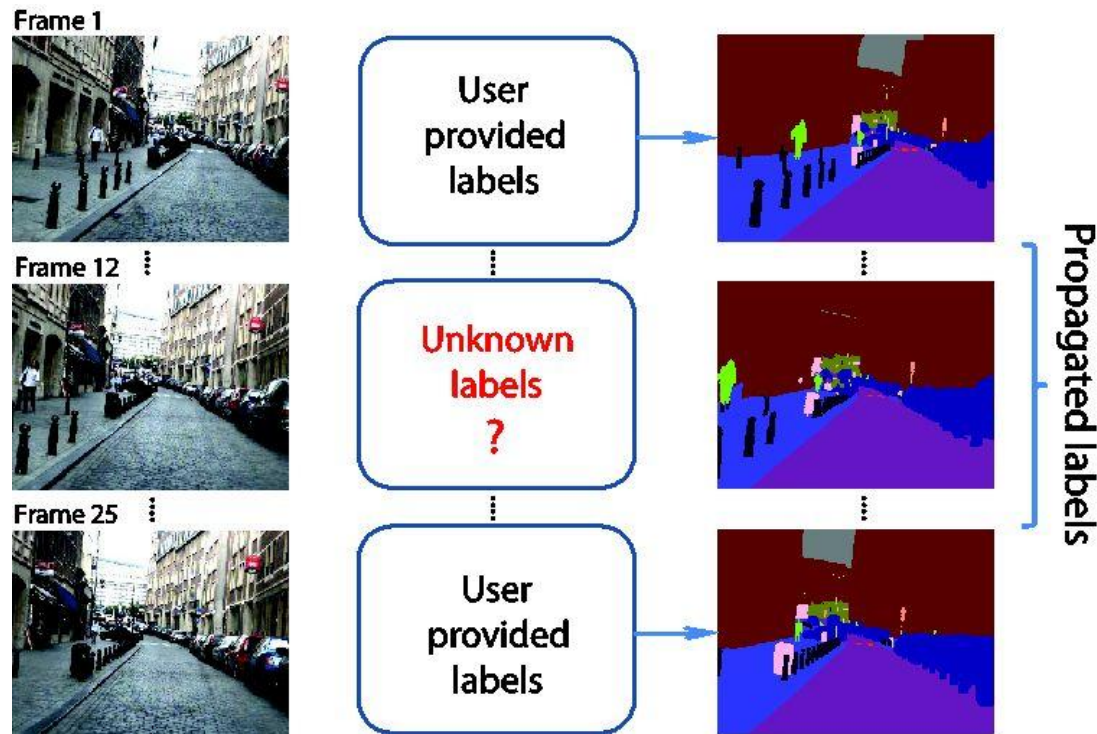
Label propagation and video analysis

CVPR'10, ICCV'11



UNIVERSITY OF
CAMBRIDGE

- Model videos with a graphical model
- First *label propagation*
 - Label non-annotated video frames automatically





max planck institut
informatik



- 2011-2014
PDoc Research Associate

Video segmentation

ICCV'11, ACCV'12, ICCV'13, ACCV'16



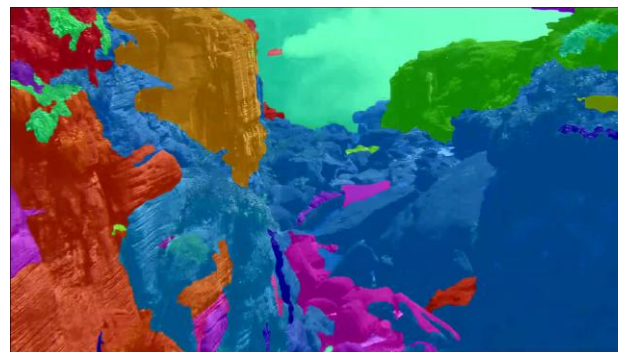
UNIVERSITY OF
CAMBRIDGE



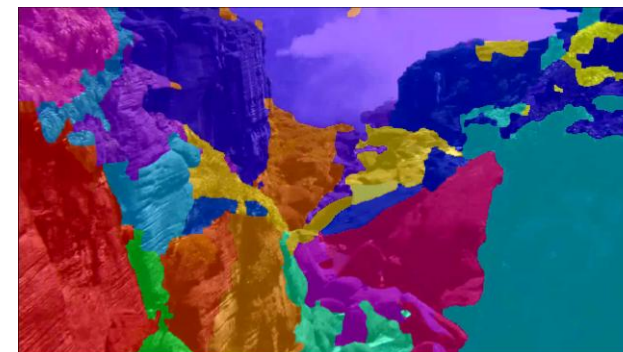
- *Learn from videos* to mimic the human perception



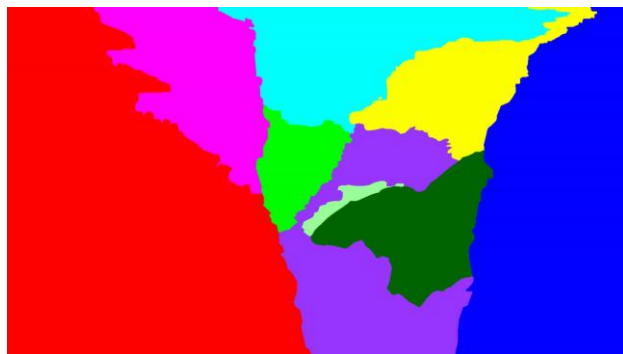
Video



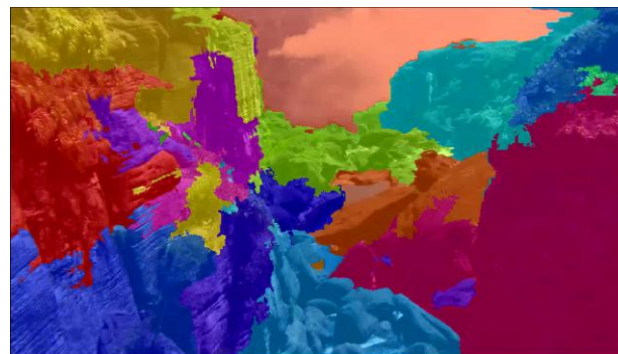
Segm. propagation



Spectral graph reduction



Ground truth



Grundmann et al. CVPR'10

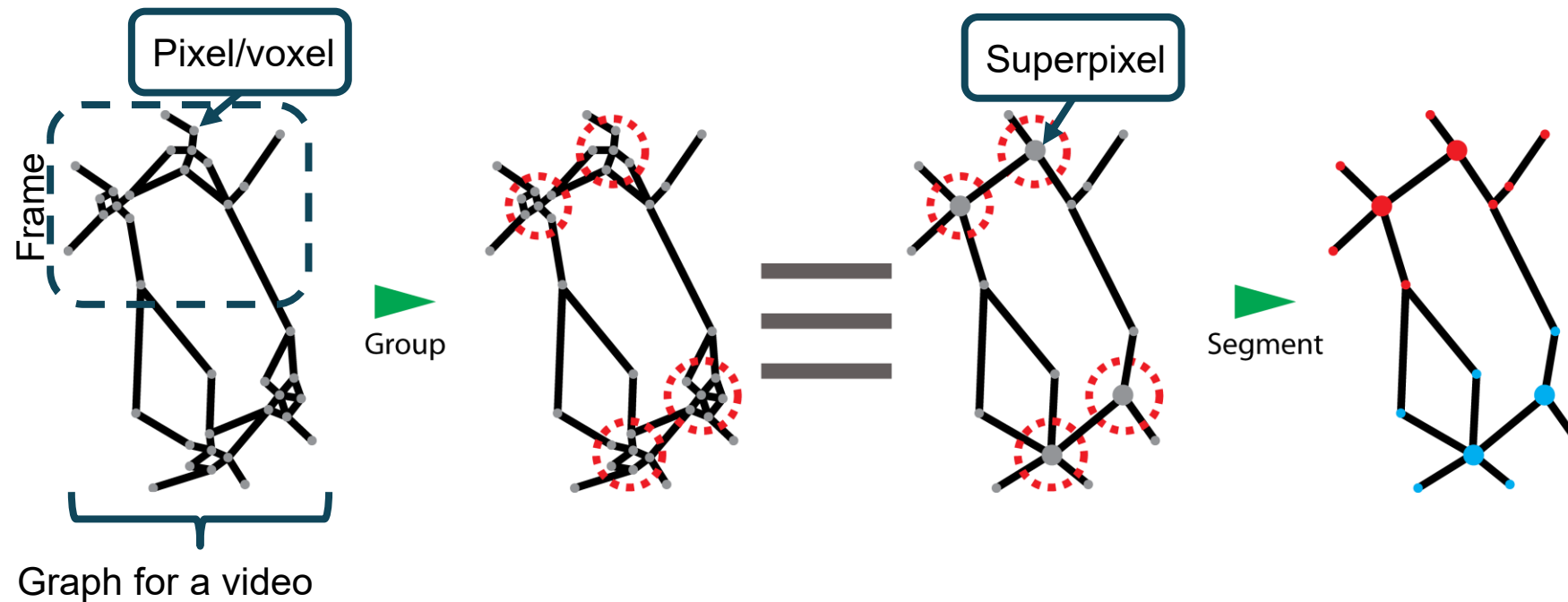


Ours

Clustering with graphs

CVPR'14, GCPR'14, CVPR'15

- Learn graph representations for videos
- Learn grouping constraints
- Prove equivalence of graph reductions



- 2014-2019
R&D Manager
Head of Computer Vision Dept



Innovation transfer

Smart office

OSRAM

- Business potential: *light is everywhere people are*



Innovation transfer

Smart office

OSRAM

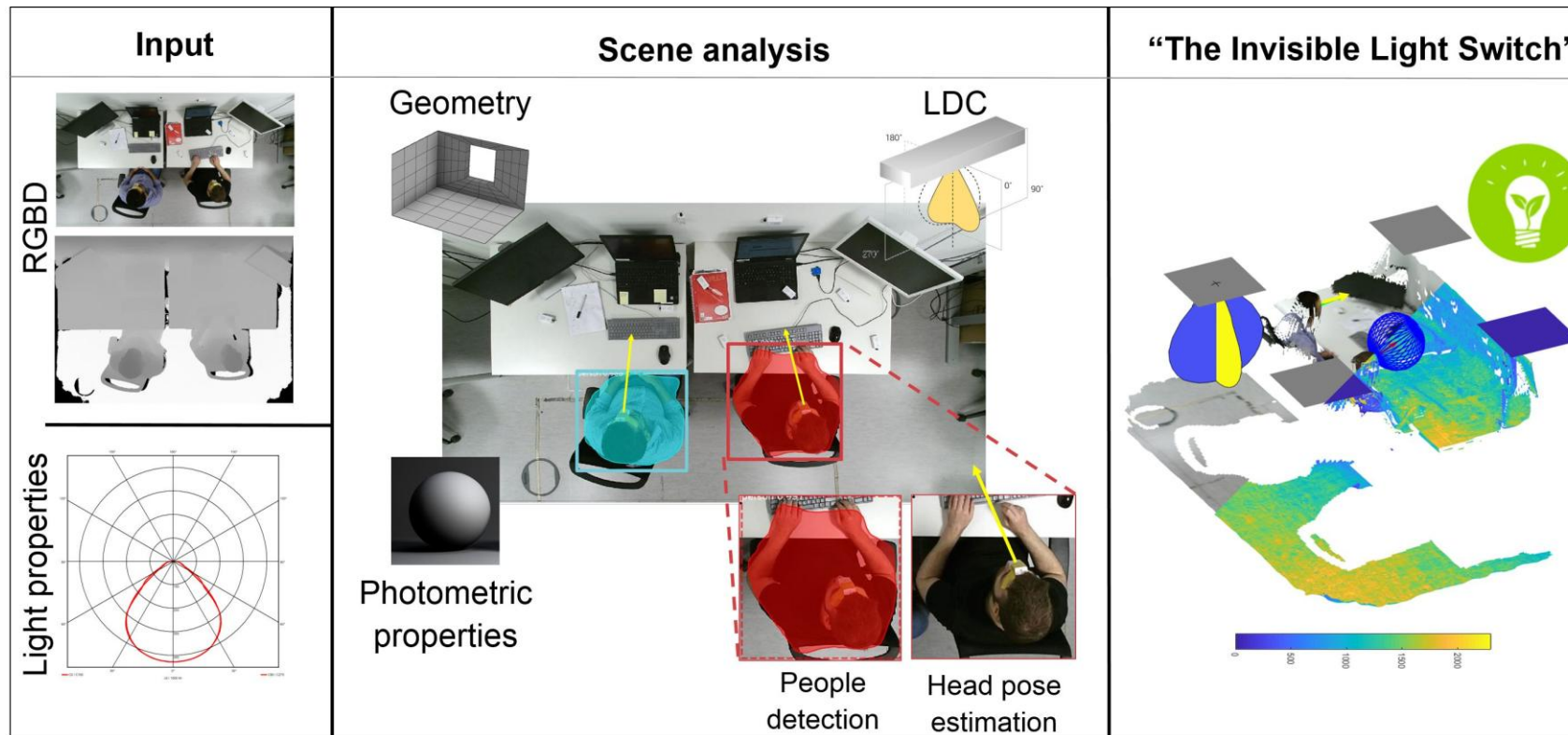
- Business potential: *light is everywhere people are*
- Objective: *make light smarter by pairing it with a camera and some intelligence*



Detection, recognition, photometry VISAPP'17, ICIP'18, 2x WACV'19

OSRAM

- Detect people from top-views and estimate their gaze
- Model lighting and estimate the perceived illumination

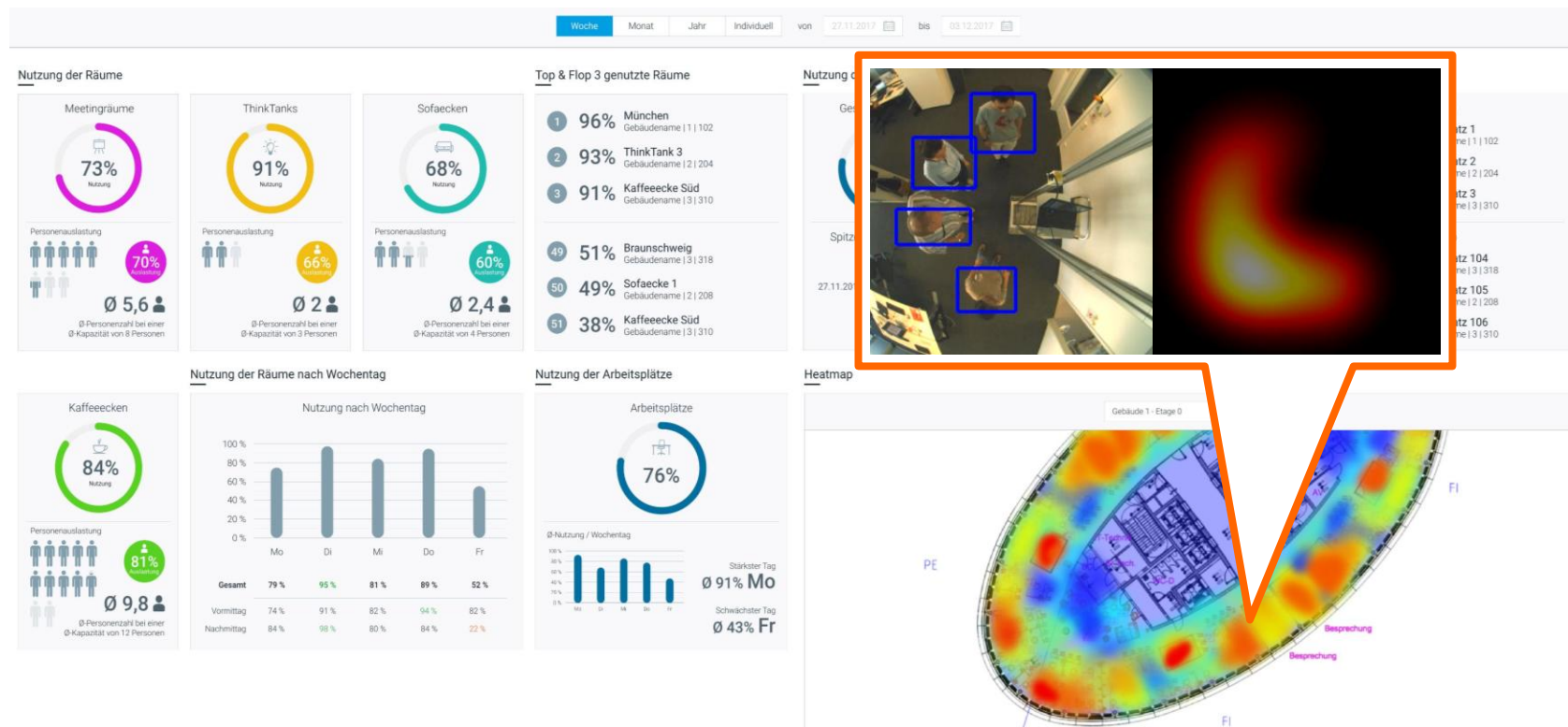


Innovation transfer

Smart office and retail



- Space utilization and people counting in offices (“visn“ spinoff)
- Customer flow and conversion rate in retail (e.g. Edeka)

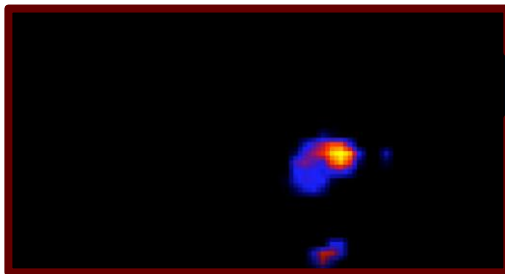


Innovation transfer

Smart office and retail

OSRAM

- Space utilization and people counting in offices (“visn” spinoff)
- Customer flow and conversion rate in retail (e.g. Edeka)
- A real product, also with thermal cameras

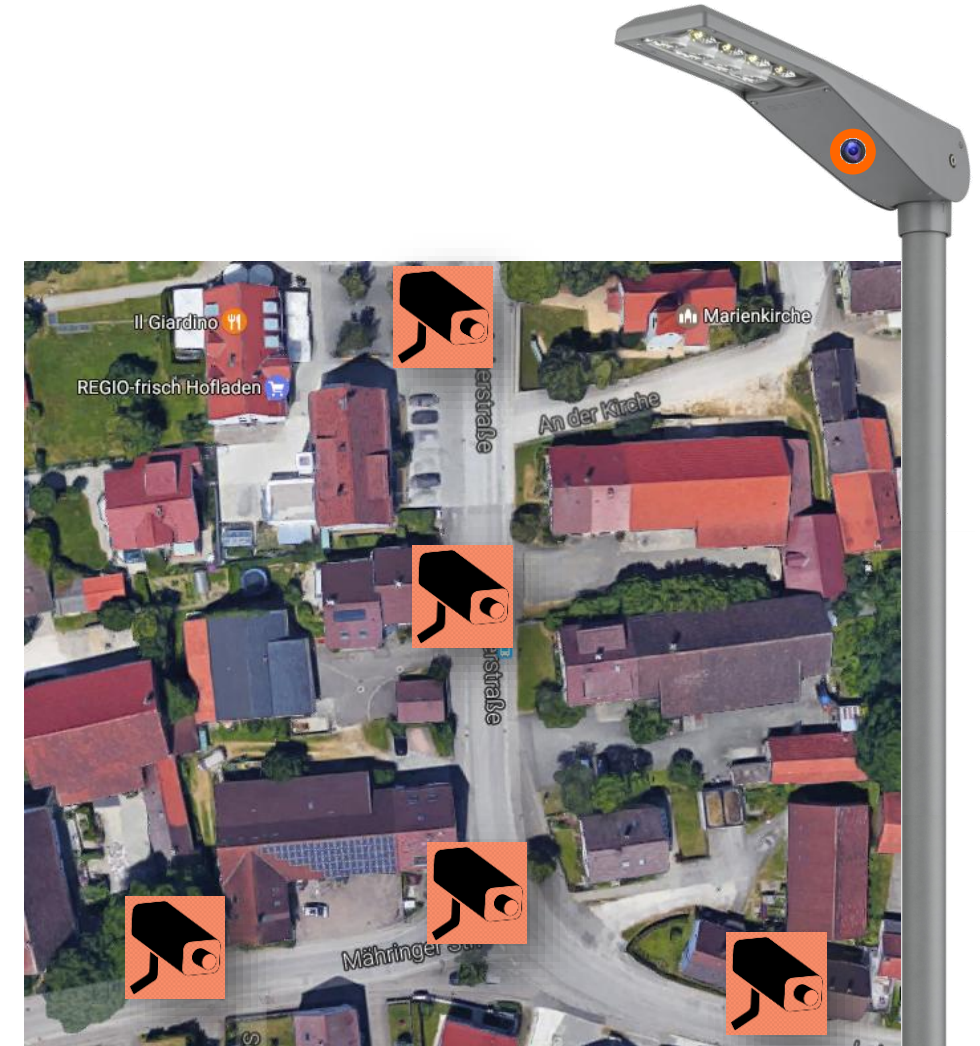


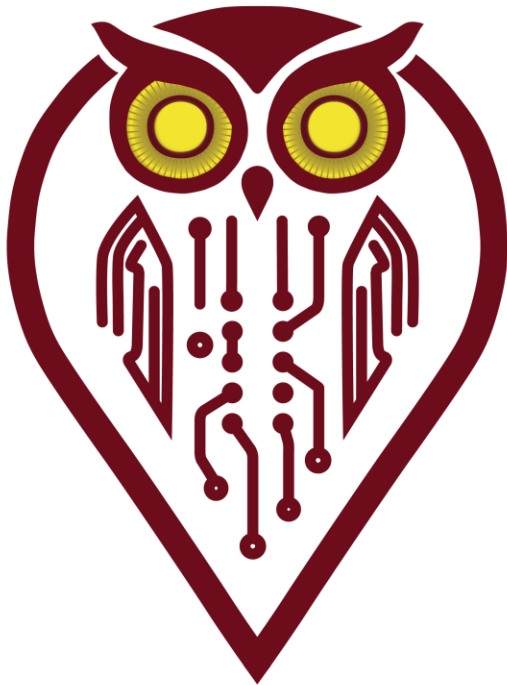
Innovation transfer

Smart city

OSRAM

- Installed in the city of Ulm
- The “city” view for autonomous driving
- With partners:





PINlab

PERCEPTION AND INTELLIGENCE

Prof. Fabio Galasso
25 September 2026



SAPIENZA
UNIVERSITÀ DI ROMA

Computer vision



Hierarchical image-text alignment

ICLR'25 (selected for oral presentation, top 2%)

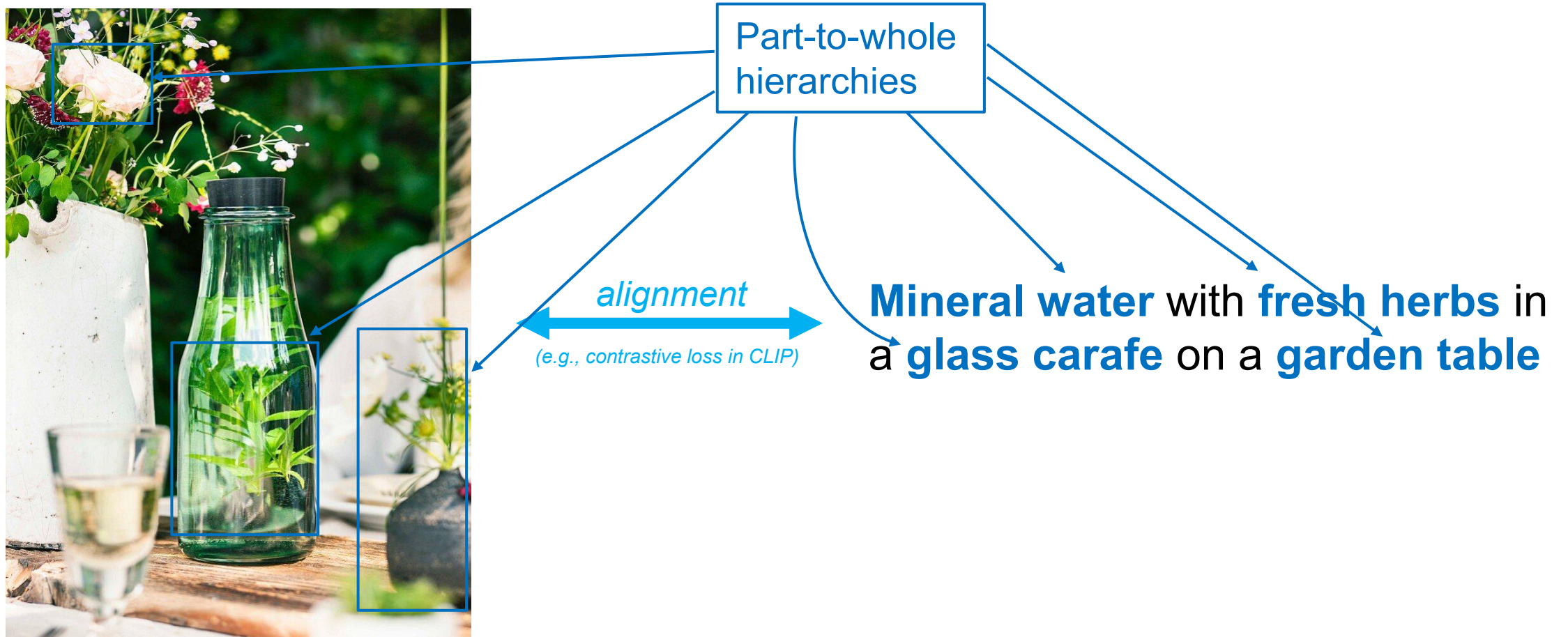
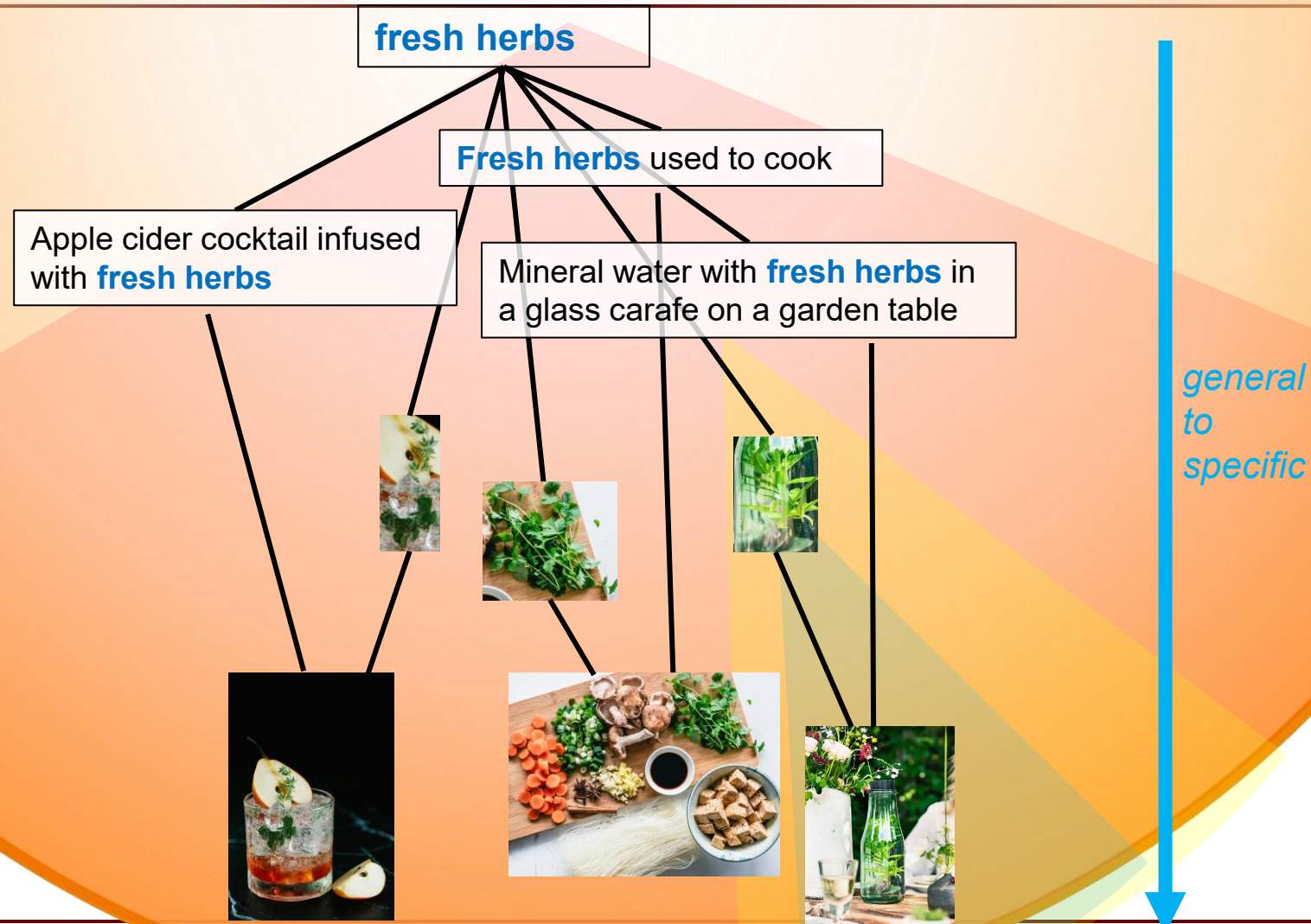


Image-text hierarchies in Hyperbolic space

ICLR'25 (selected for oral presentation, top 2%)



Text, Motion and Scene understanding and retrieval

ICCV'25

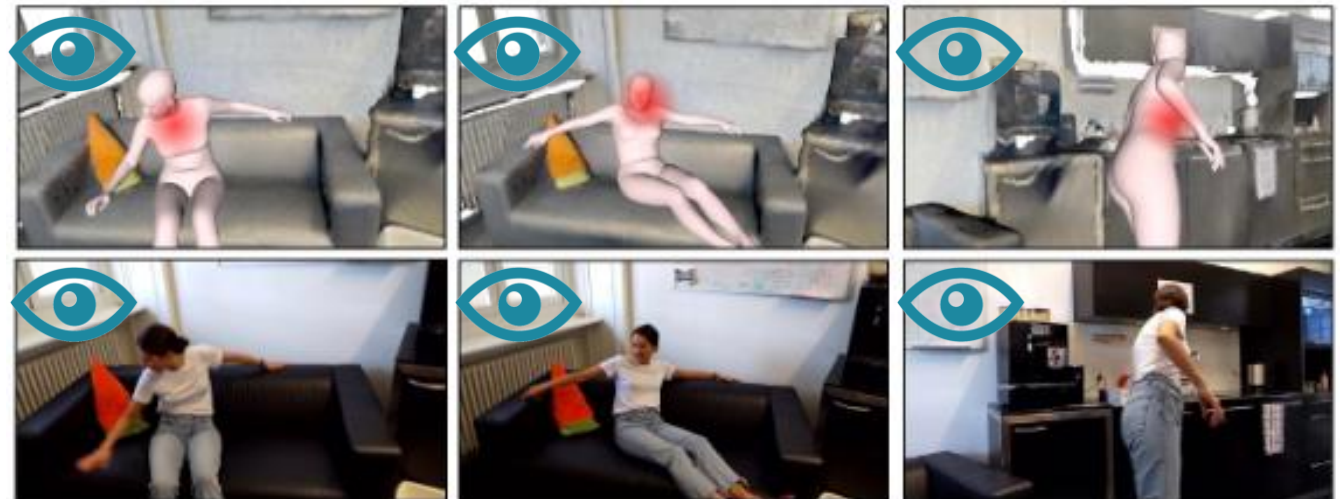
- Understand by jointly reasoning on human motion + text + scene



SEE-ME Social Ego-Mesh Estimation

WACV'25

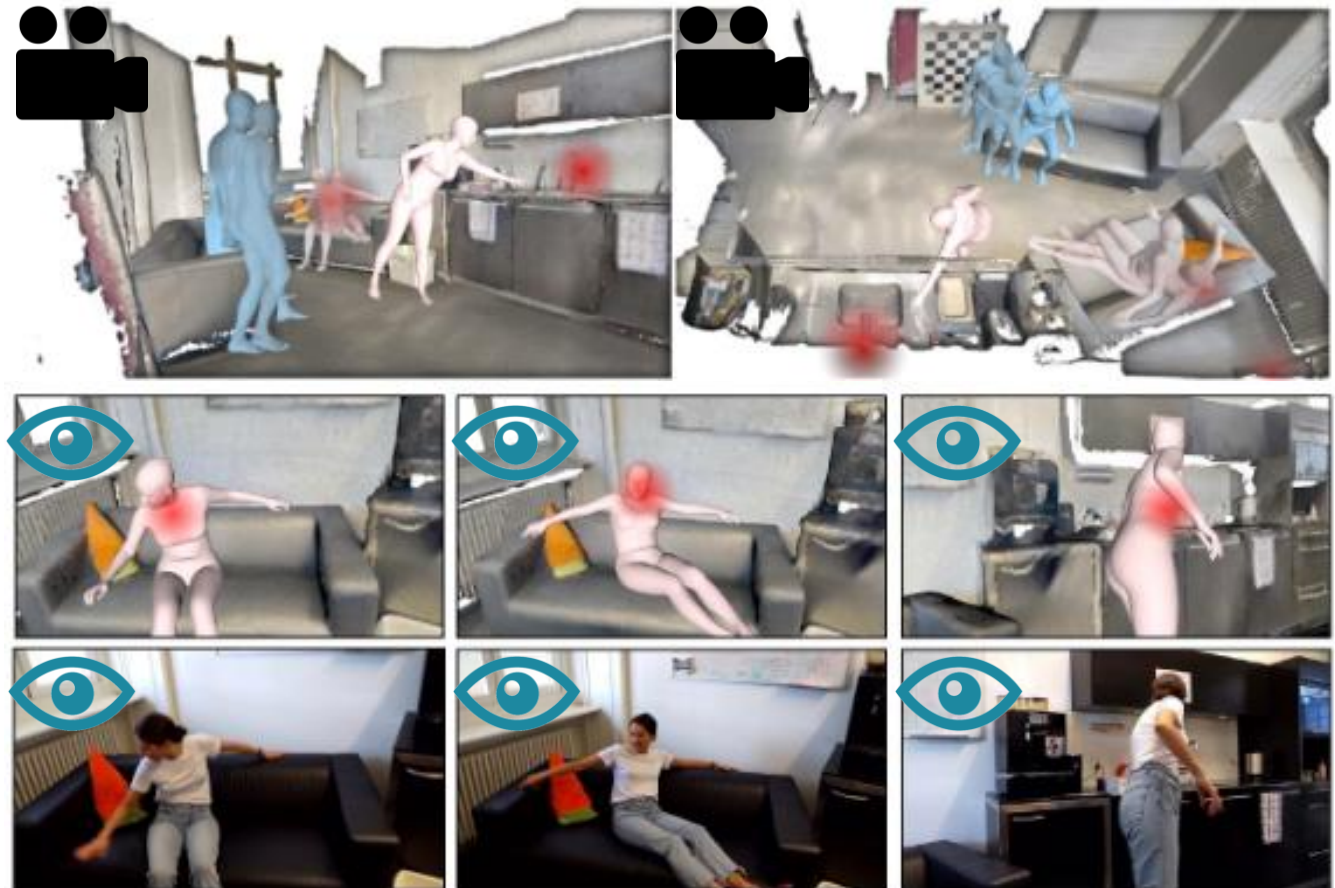
- Consider the **interactee**'s motion and the **scene**



SEE-ME Social Ego-Mesh Estimation

WACV'25

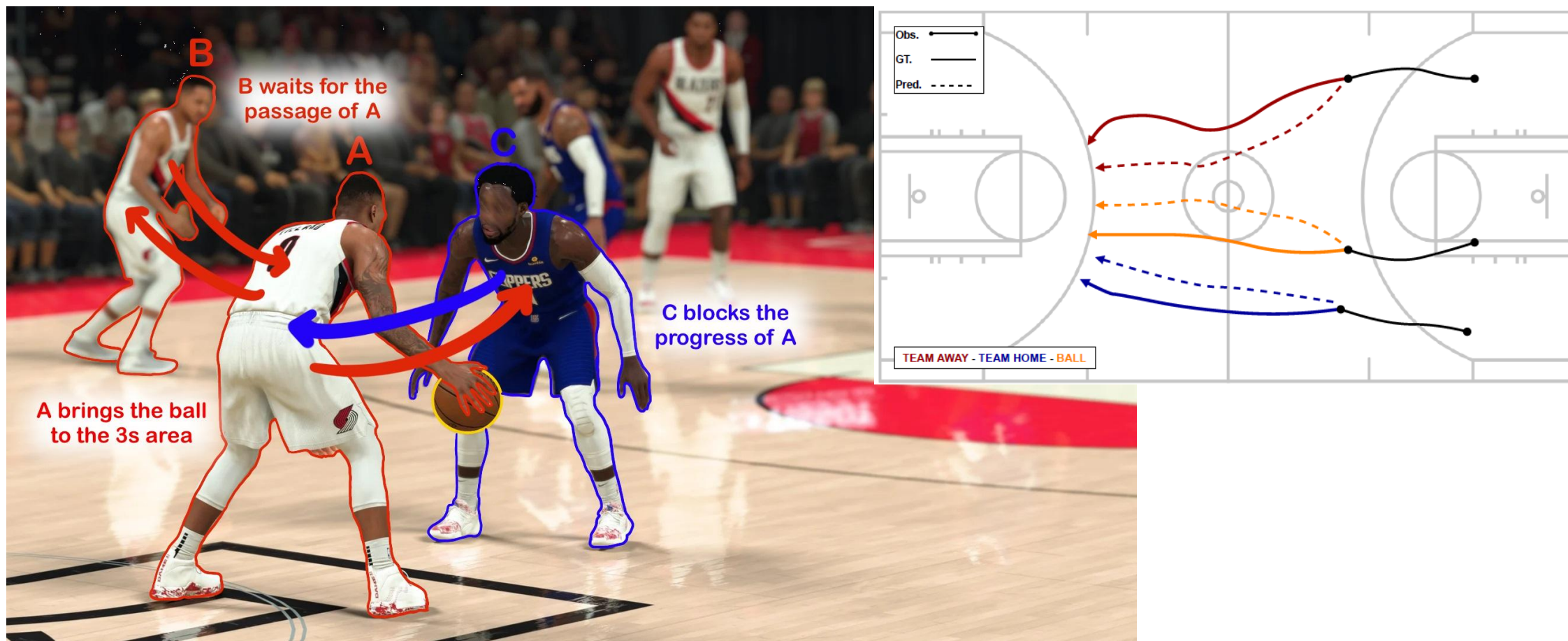
- Consider the **interactee**'s motion and the **scene**
- Reconstruct the **viewer**'s motion



About latent roles in forecasting players in team sports

ICLR'23 wks, Neural Processing Letters'24

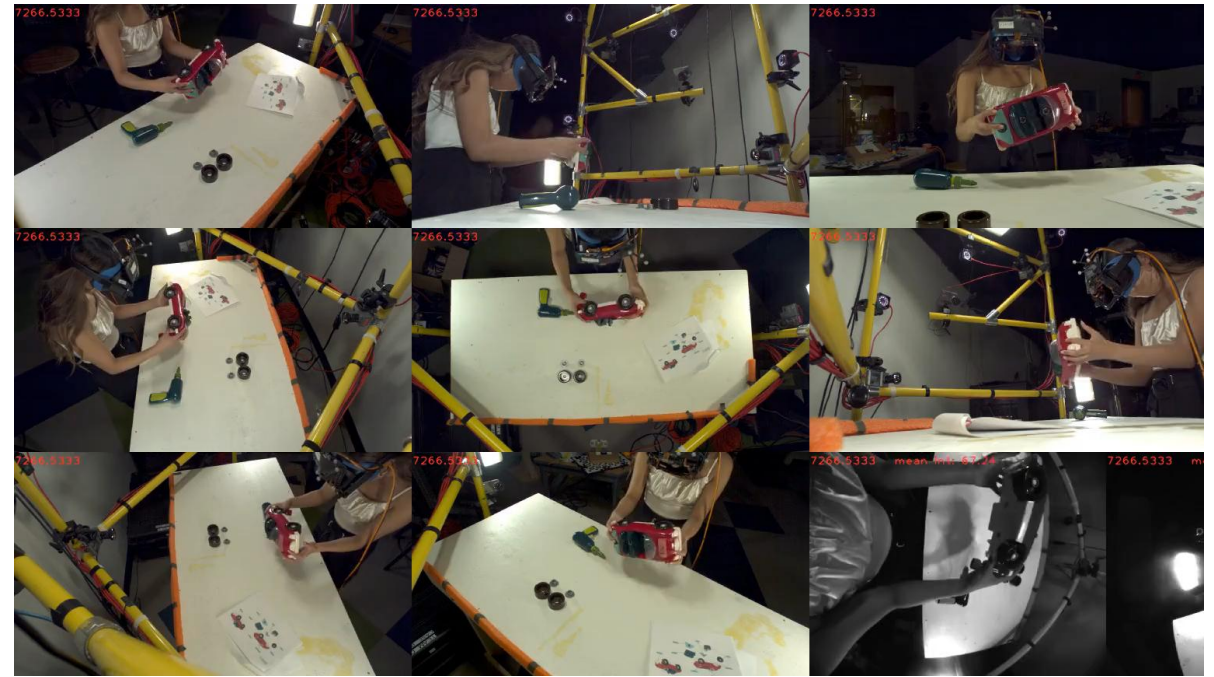
- Learn role-based interaction between basketball players



(Video) Anomaly Detection

ICCV'23, CVPR'24, PR'24

- Anomalous behavior, accidents, fights
- Anomalous procedures



Robotics



Source: <https://www.youtube.com/watch?v=6zPvT0ig1VM>

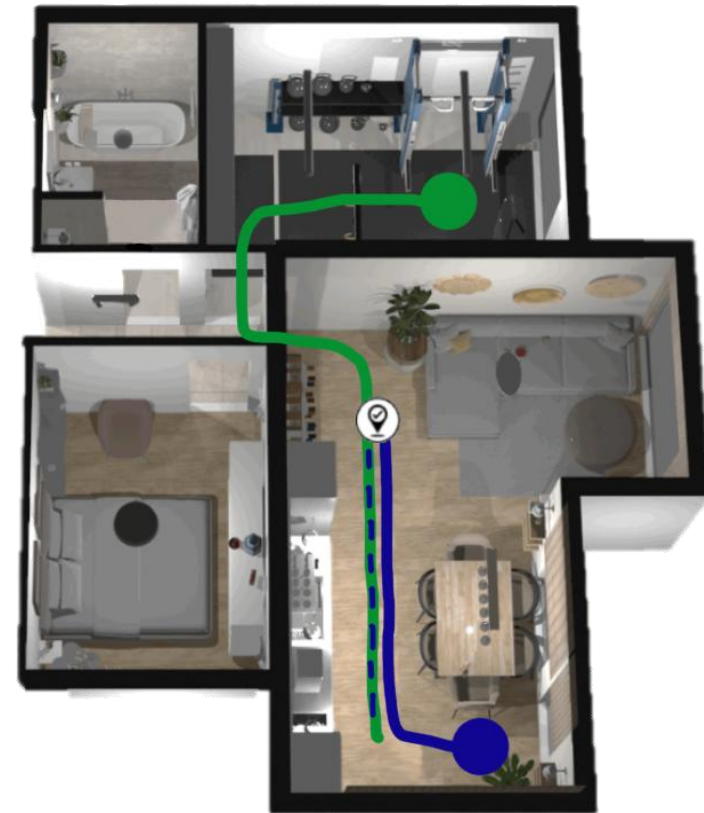
Robot perception and interaction for social navigation

ICLR'25 (selected as Highlight, top 5%), IROS'26

- Find and follow

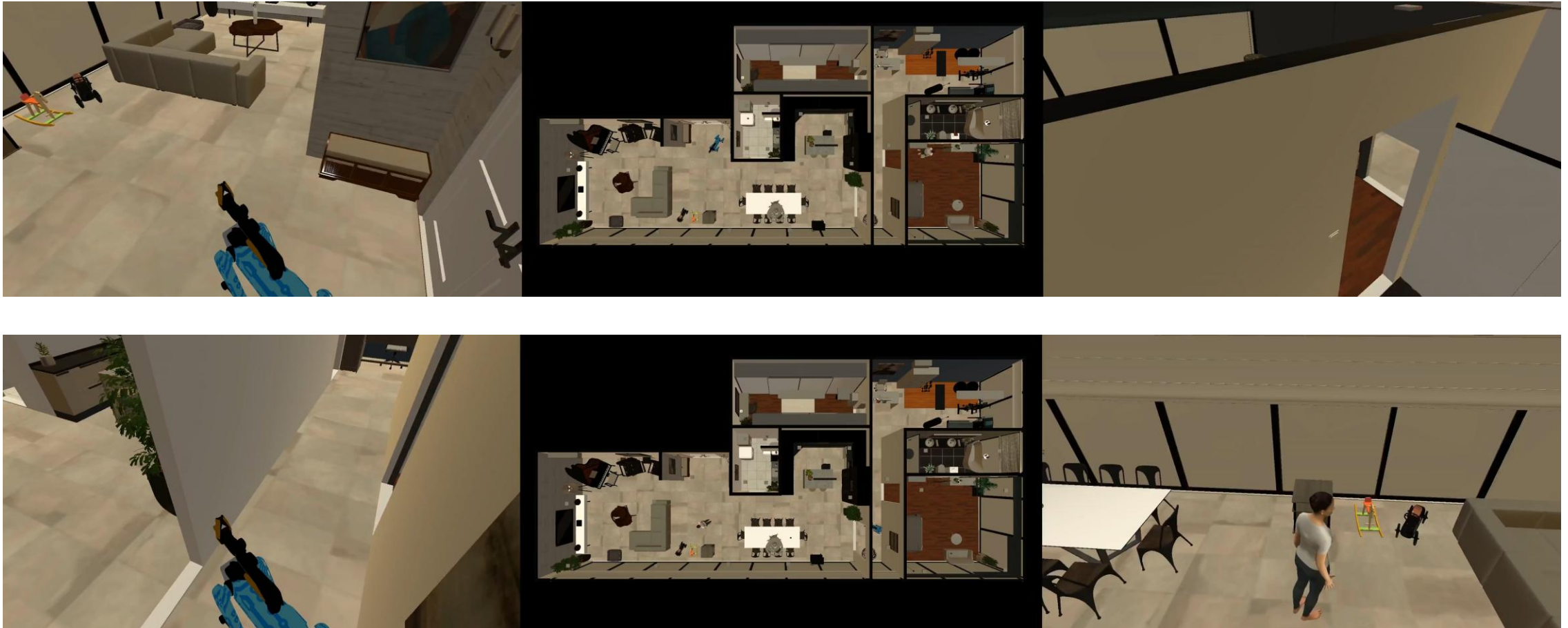


- Humanoid start position
- Robot start position
- 📍 Humanoid found



Robot perception for social navigation

ICLR'25 (selected as Highlight, top 5%), IROS'26



Robot perception and interaction for social navigation

ICLR'25 (selected as Highlight, top 5%), IROS'26

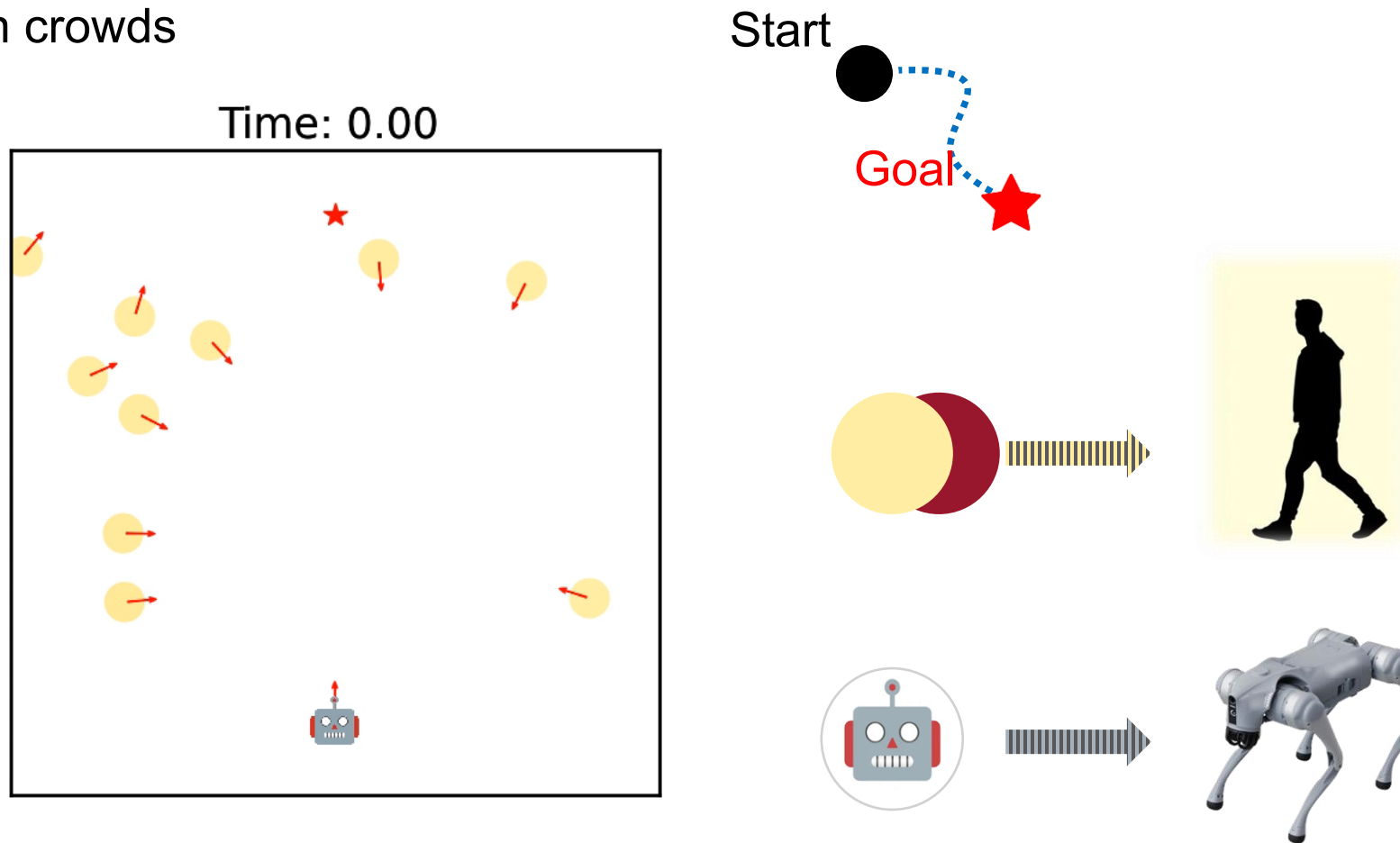
- Interact and locate with verbal cues



Robot perception for social navigation

IROS'24 (*selected for oral presentation*)

- Robots in crowds

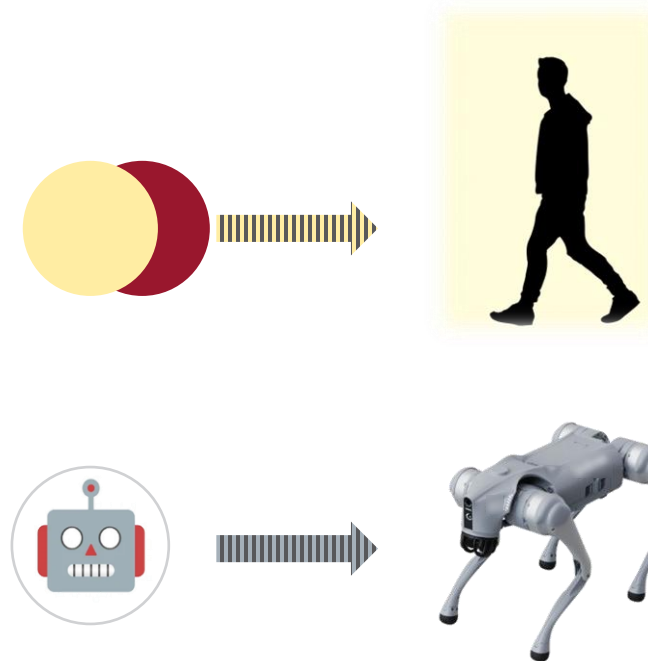
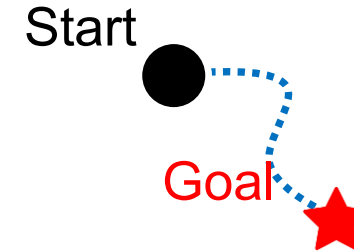
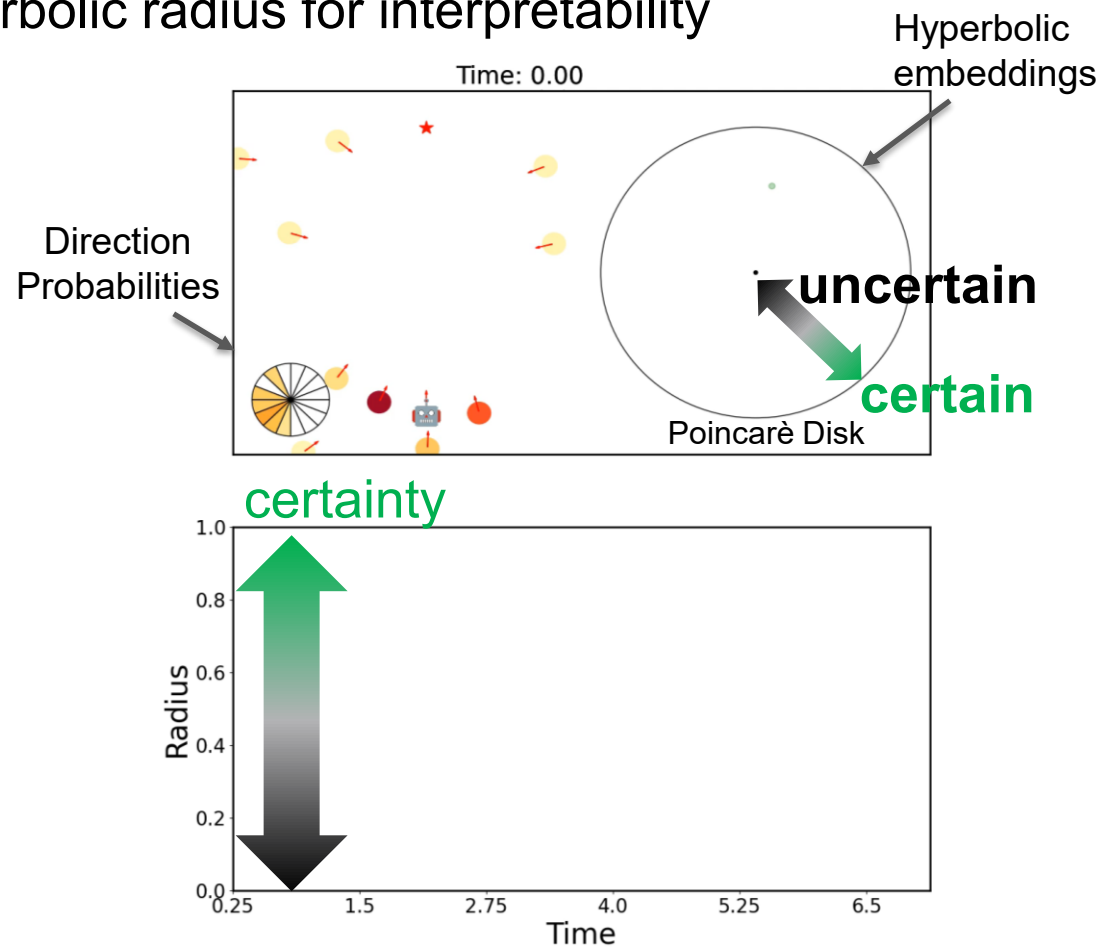


Images courtesy of: Automata, <https://www.generationrobots.com/it/404127-robot-quadrupede-go2-edu-plus.html>, <https://www.publicdomainpictures.net/it/view-image.php?image=70565>

Robot perception for social navigation

IROS'24 (*selected for oral presentation*)

- Hyperbolic radius for interpretability

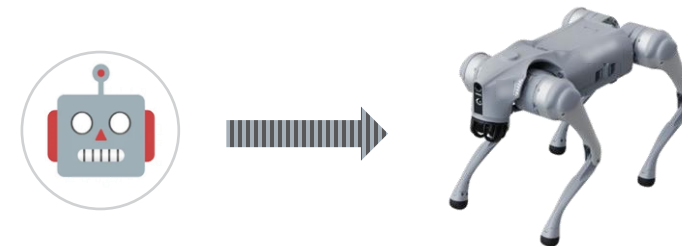
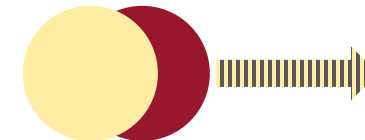
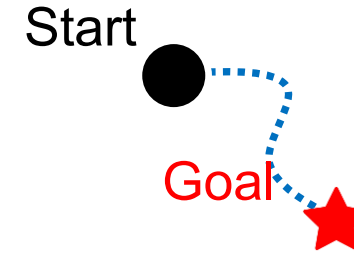
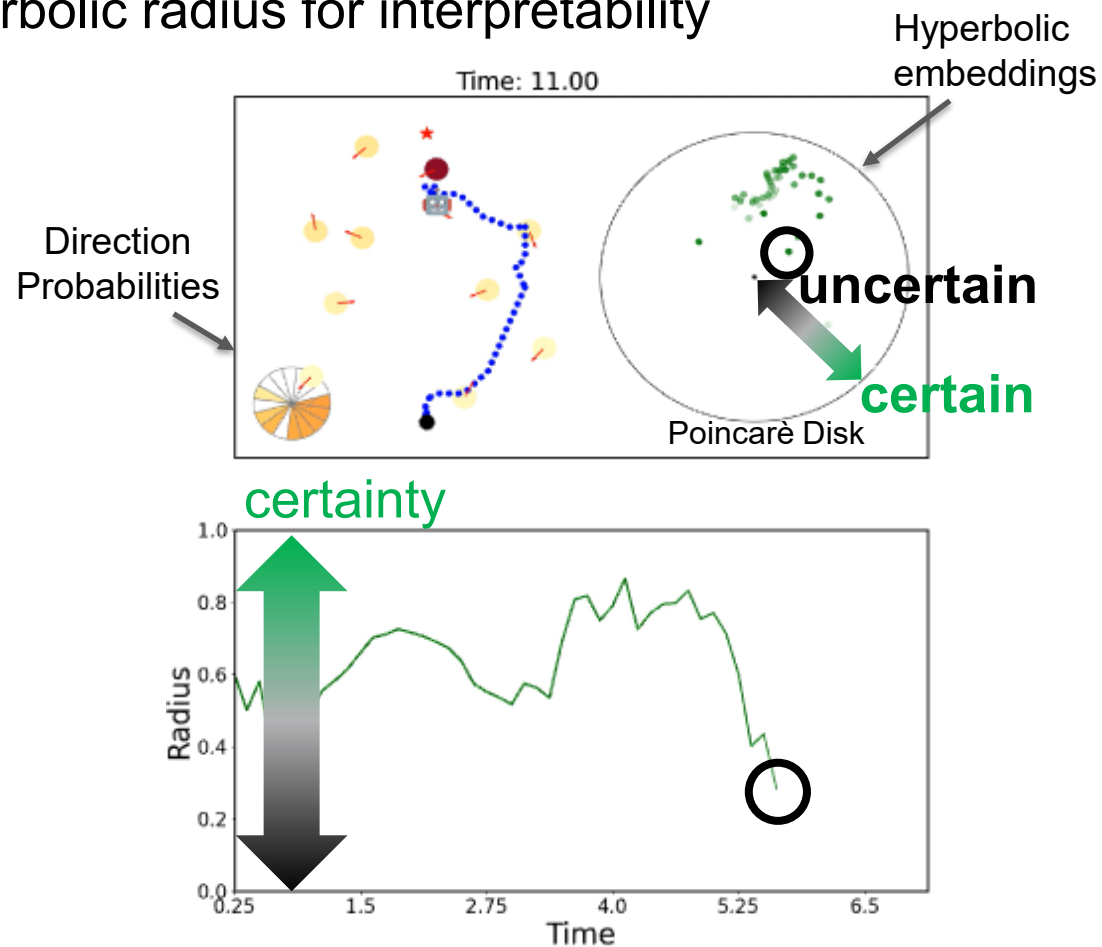


Images courtesy of: Automata, <https://www.generationrobots.com/it/404127-robot-quadrupede-go2-edu-plus.html>, <https://www.publicdomainpictures.net/it/view-image.php?image=70565>

Robot perception for social navigation

IROS'24 (*selected for oral presentation*)

- Hyperbolic radius for interpretability

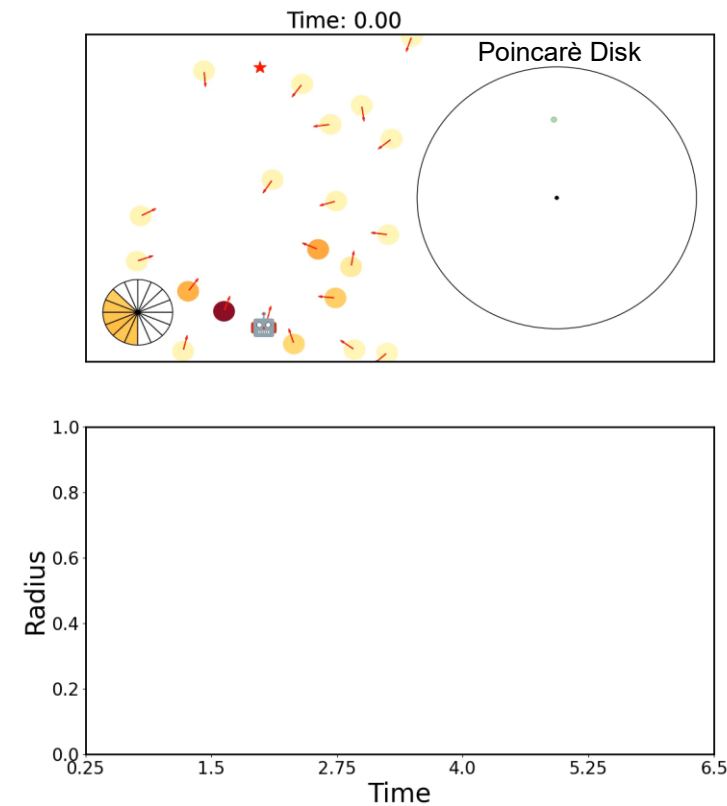
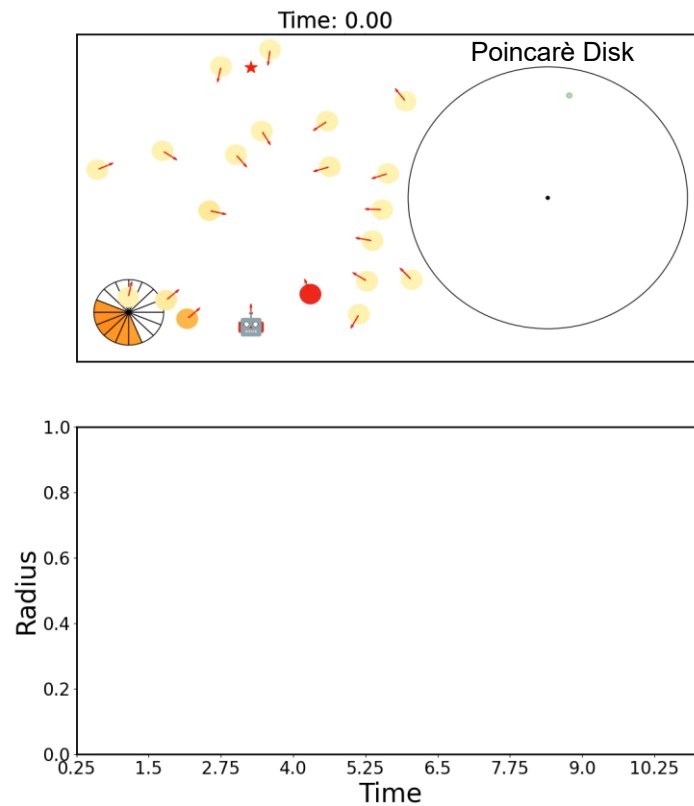


Images courtesy of: Automata, <https://www.generationrobots.com/it/404127-robot-quadrupede-go2-edu-plus.html>, <https://www.publicdomainpictures.net/it/view-image.php?image=70565>

Robot perception for social navigation

IROS'24 (*selected for oral presentation*)

- Hyperbolic radius for interpretability



Human Pose Forecasting for Human-Robot Interaction

ICCV'21, ECCV'22

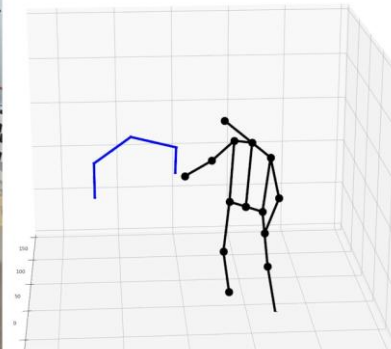
- Endow robots with human future motion and intentions



Human Pose Forecasting for Human-Robot Interaction

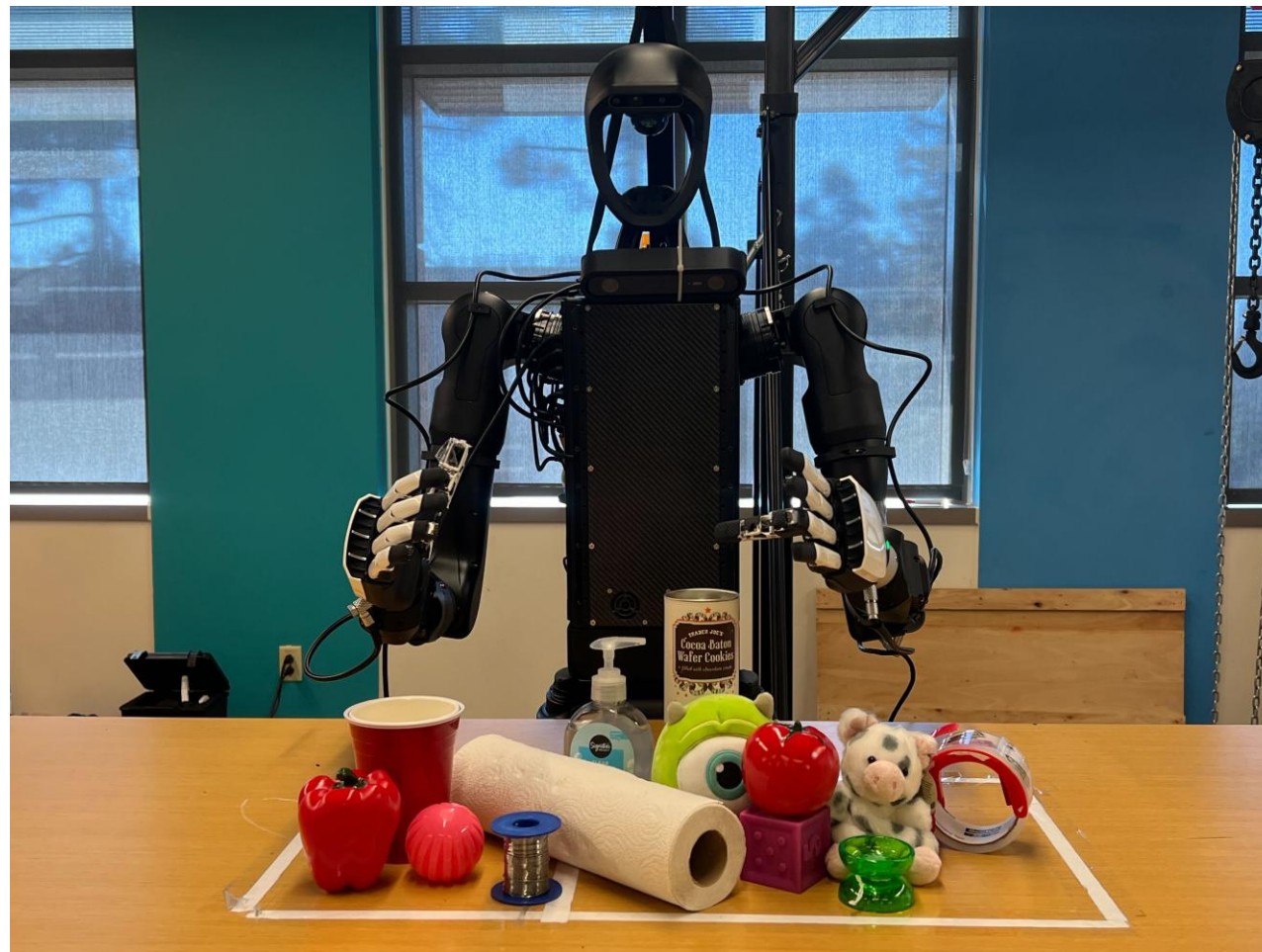
ICCV'21, ECCV'22

- Endow robots with human future motion and intentions
- Collision-free interaction within shared environments



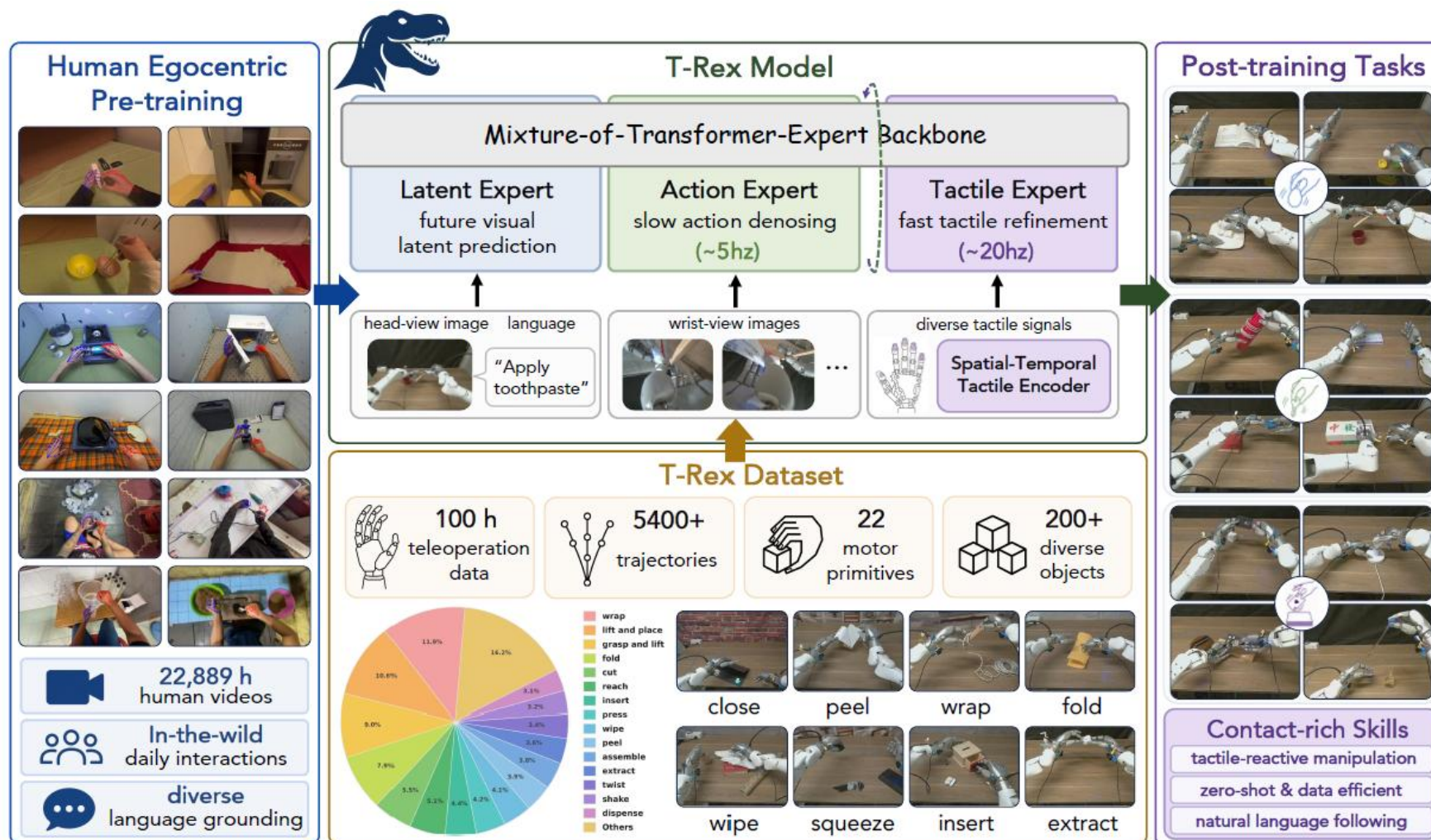
Robotic manipulation and humanoid control

ICLR'26



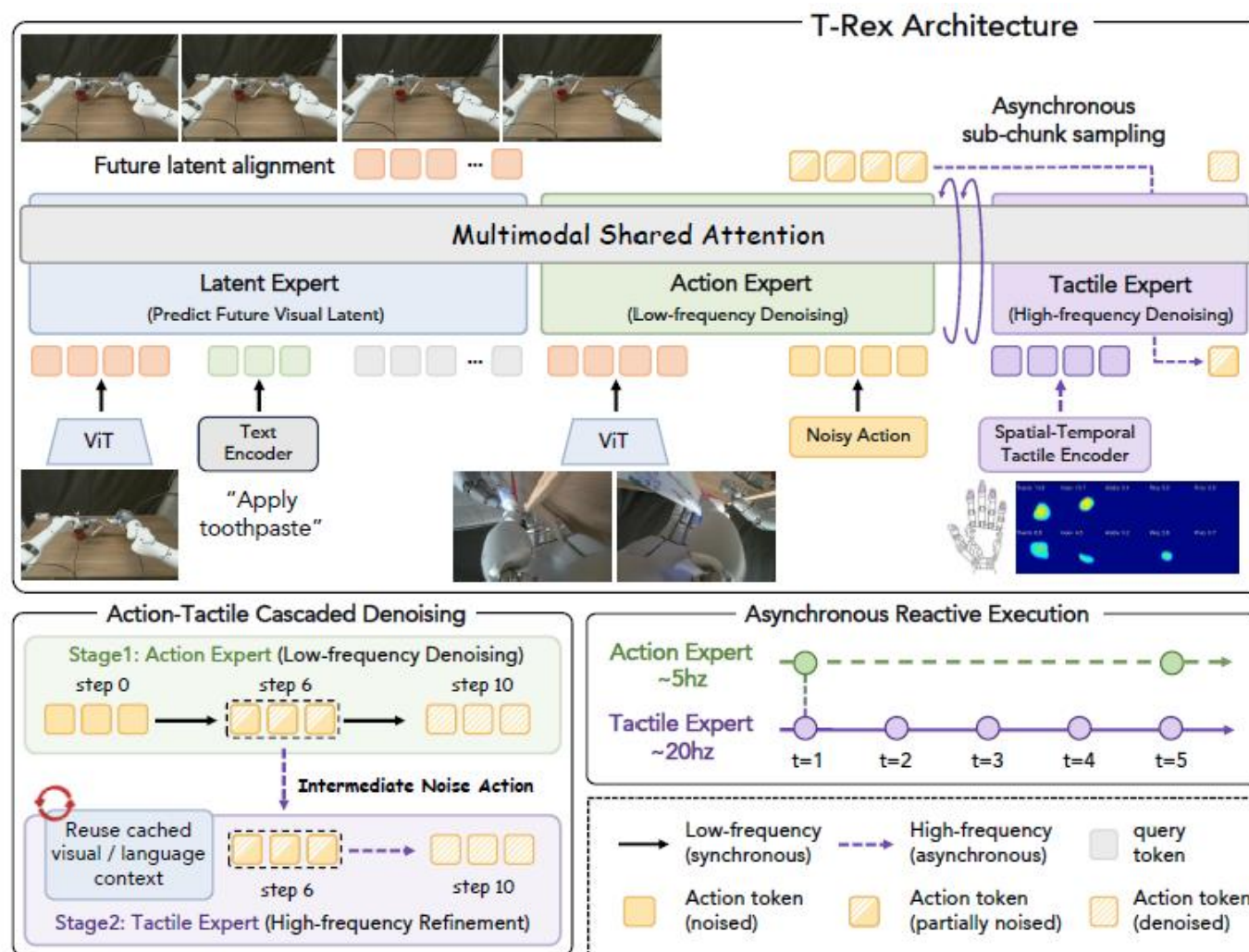
T-Rex: Tactile-Reactive Dexterous Manipulation

CoRL'26



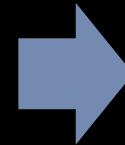
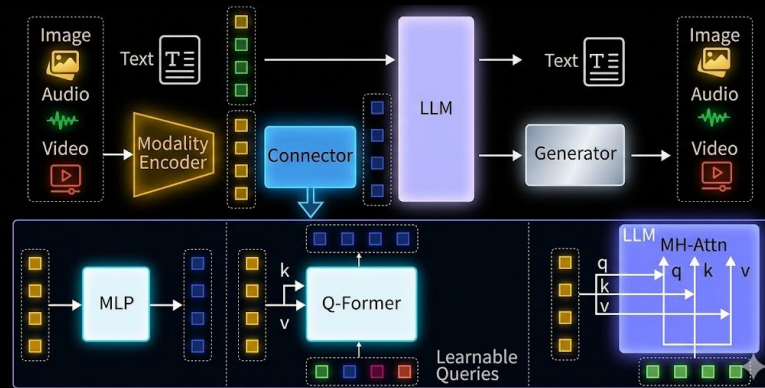
T-Rex: Tactile-Reactive Dexterous Manipulation

CoRL'26



Foundation models for text, video, etc.

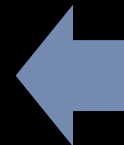
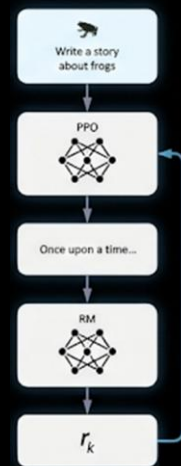
Architecture
(LLM, VLM,
MMLM)



Self-Supervised
pre-training



Self-improvement
(RLHF, RLAI, RLVR)



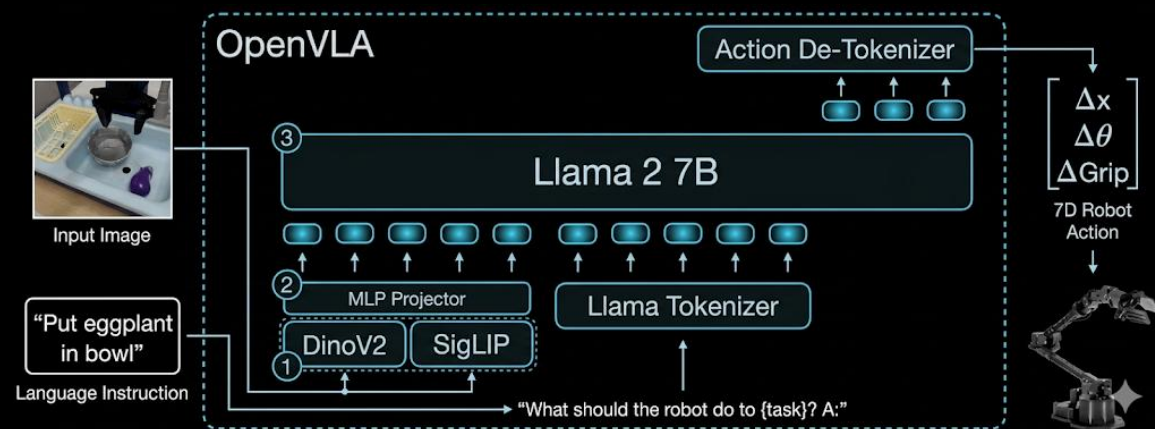
written by
people:

<USER>
Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>
"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions...

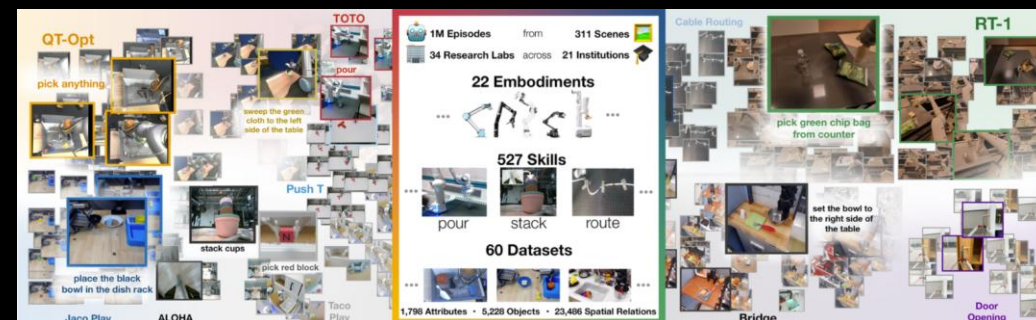
Supervised
fine-tuning

Towards robot foundational models



Architecture (VLA)

Large-scale pre-training



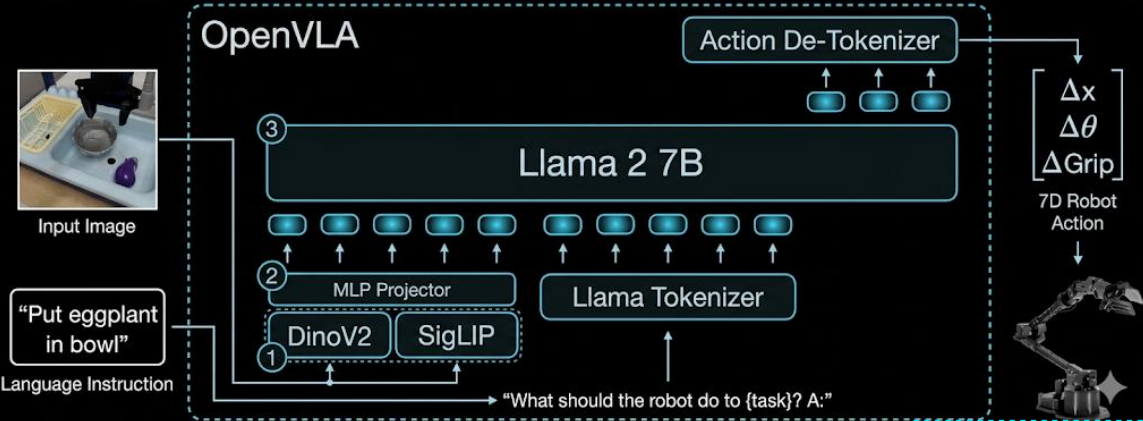
Self-improvement
(online RL)



Can we skip collecting
demonstrations and action
labels for every new task?

Task-specific
fine-tuning

Towards robot foundational models



Architecture (VLA)



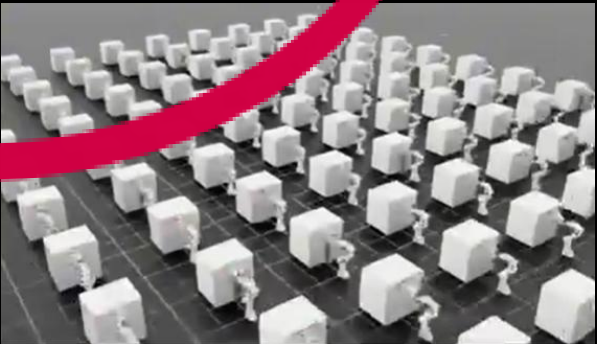
Large-scale pre-training



Self-improvement
(online RL)

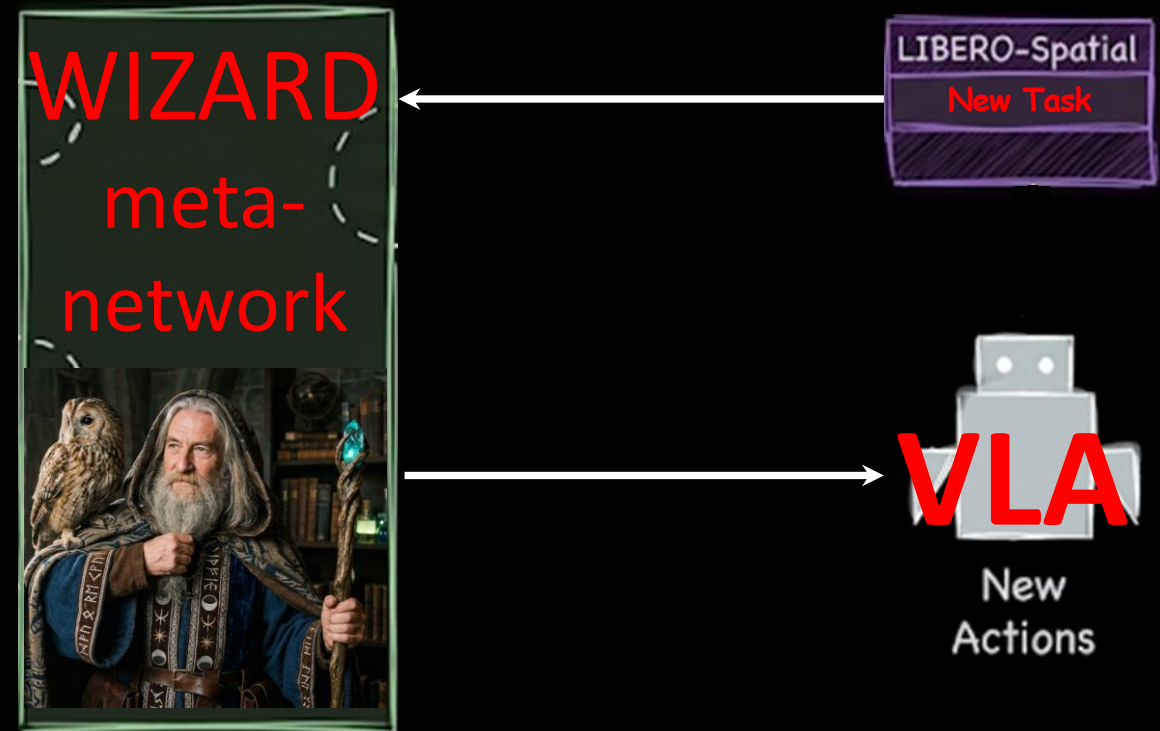


Task-specific
fine-tuning



Meta-Learning for VLAs

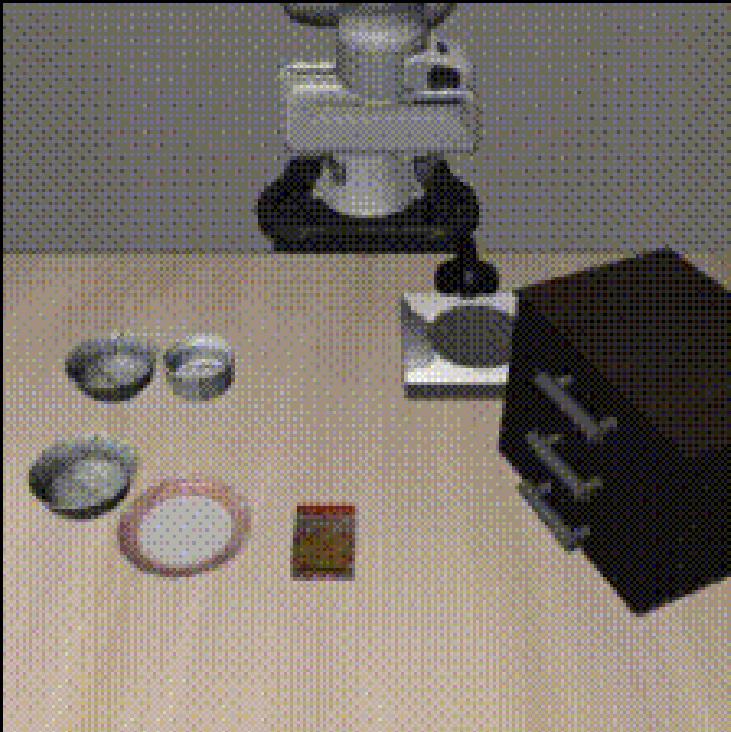
Task-specific VLA by estimating LoRA adapters from one demonstration w/o action labels



The WIZARD paradigm



Task-specific VLAs from a single demonstration (no action labels)



LIBERO: Spatial task

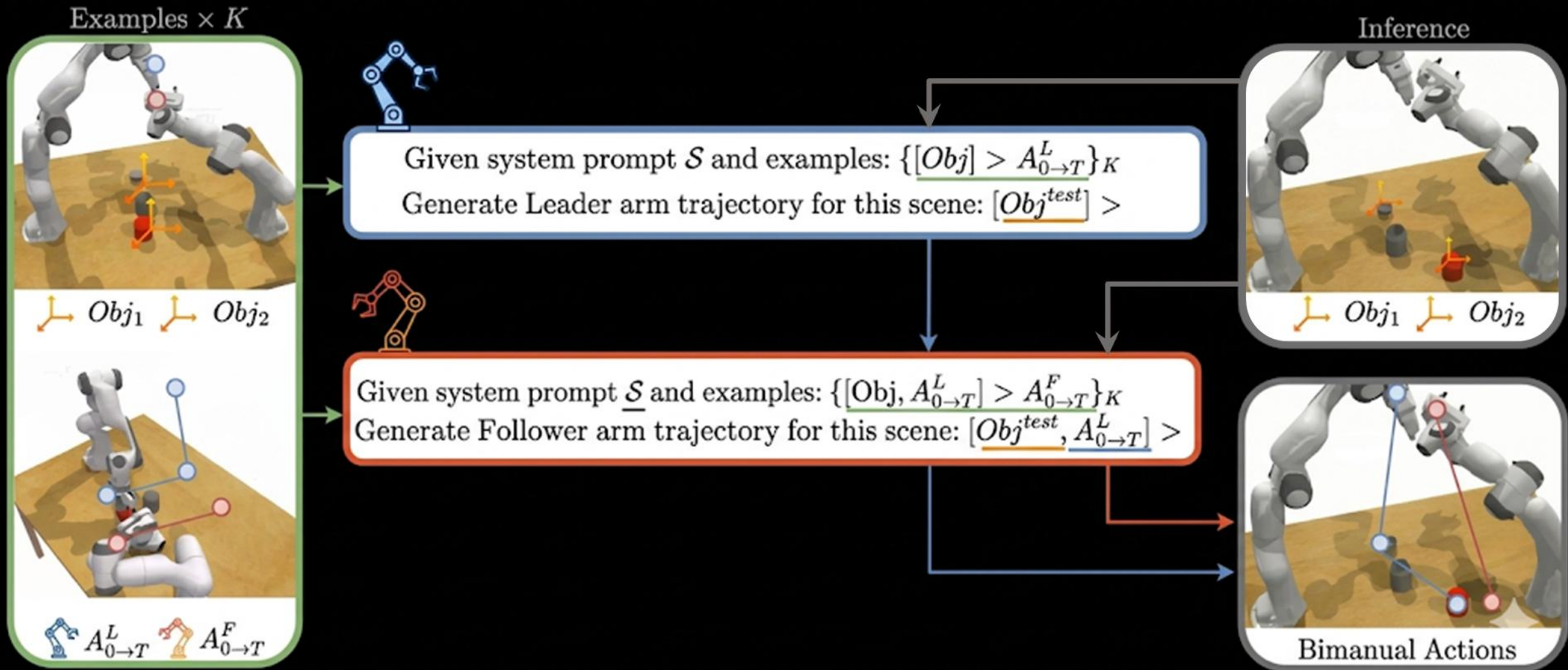


Object task

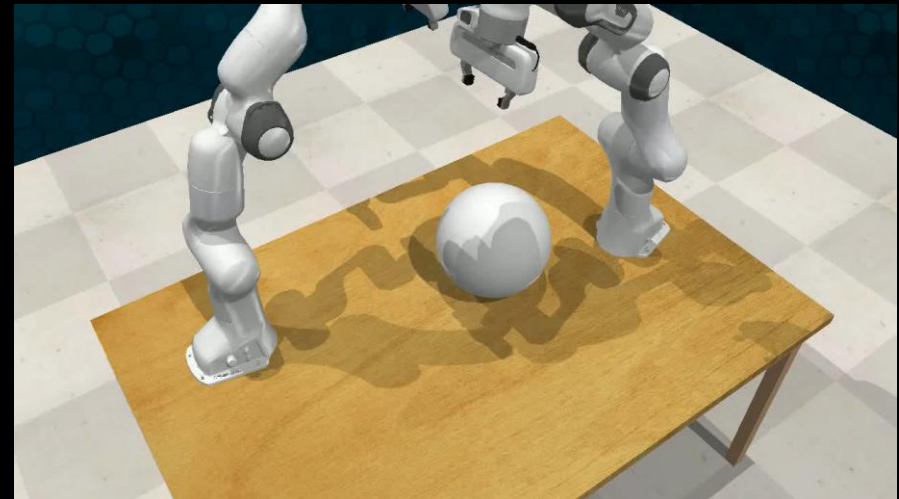
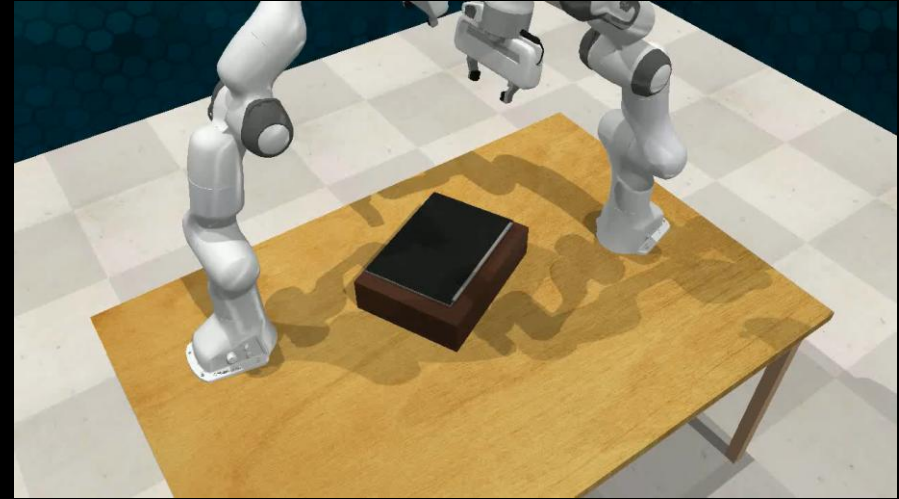
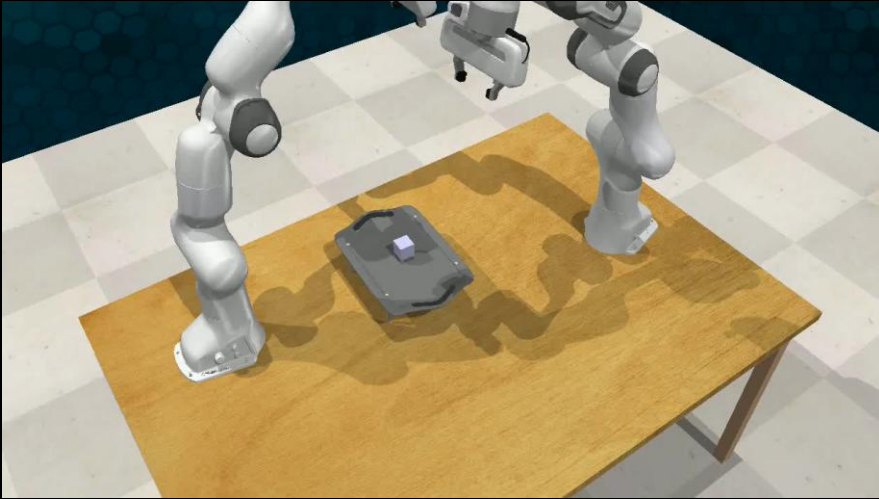


Goal task

LLMs for bimanual manipulation

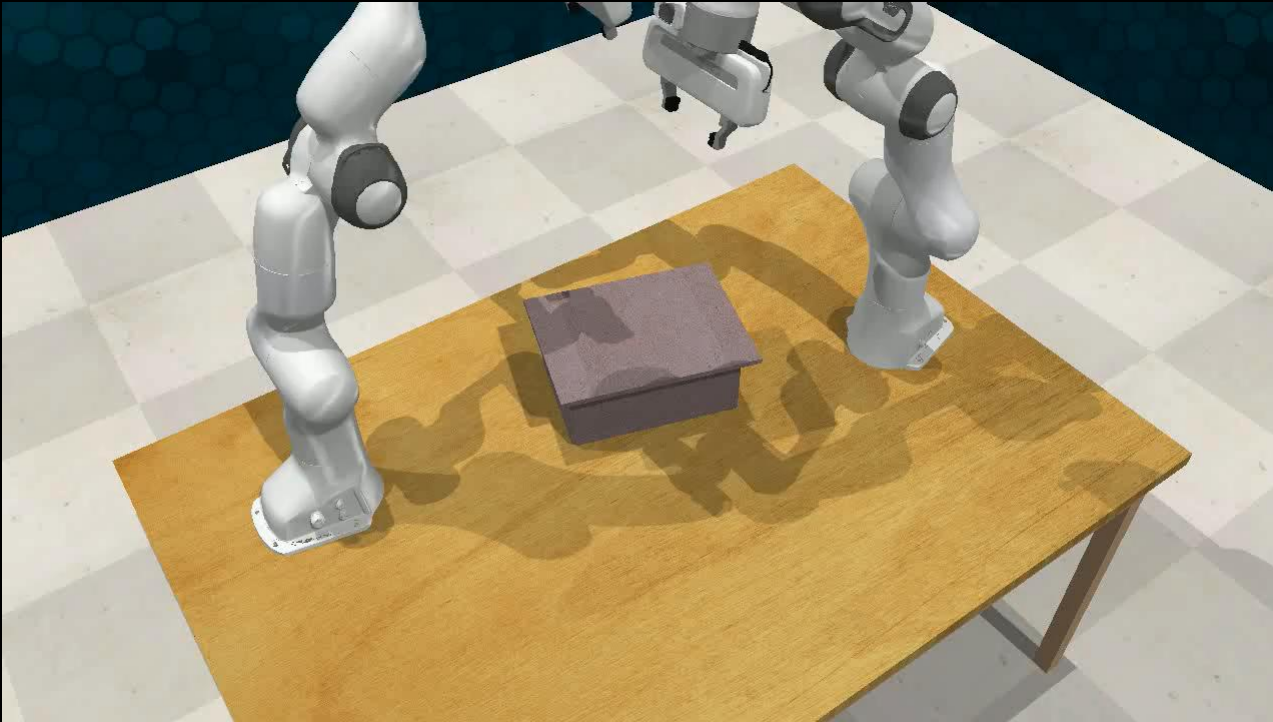


LLMs for bimanual manipulation



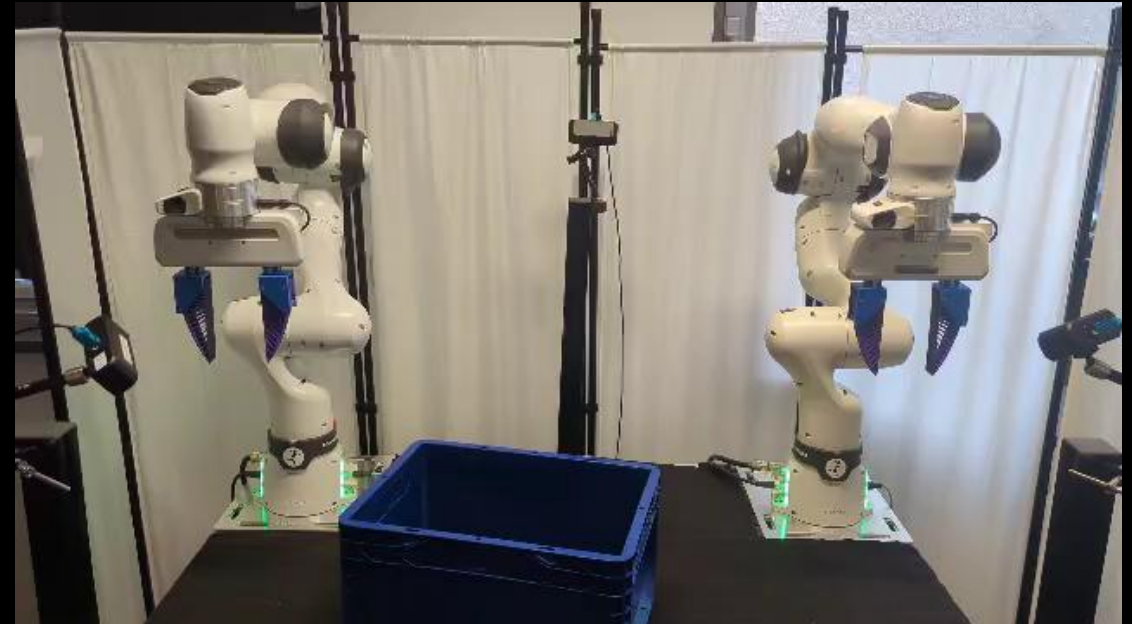
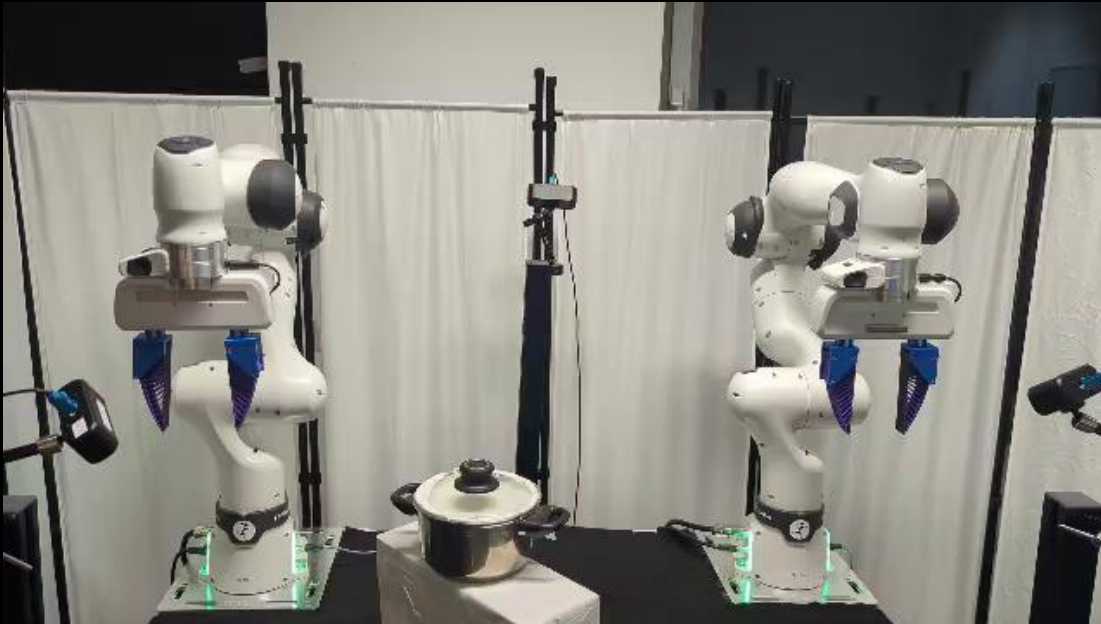
LLMs for bimanual manipulation

Our (zero-shot) models surpasses the state-of-the-art (finetuned) VLA, 3DFA [Gkanatsios et al. ArXiv'25], on new actions



LLMs for bimanual manipulation

Real-world test with Franka Panda arms



DILLO

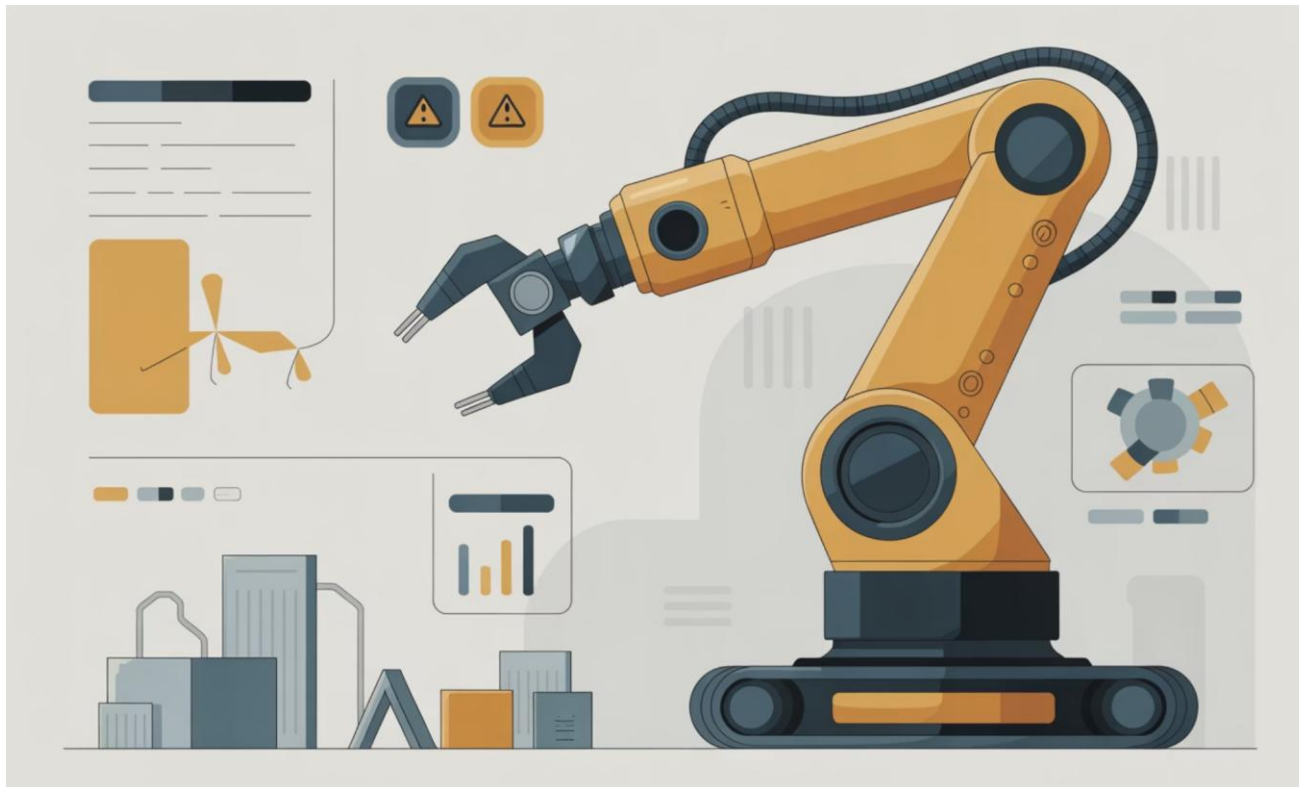
Describe-Then-Act: Proactive Agent Steering via Distilled Language-Action World Models

Massimiliano Pappa, Luca Romani, Valentino Sacco, Alessio Palma,
Stéphane Lathuilière, Fabio Galasso, Xavier Alameda-Pineda, Indro Spinelli



ECCV'26

Foresight vs. Latency Dilemma

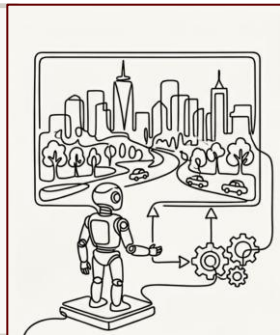


Reactive (Post-Hoc)

Diagnoses failures *after* they occur —
cannot prevent irreversible actions

Proactive Visual Sim

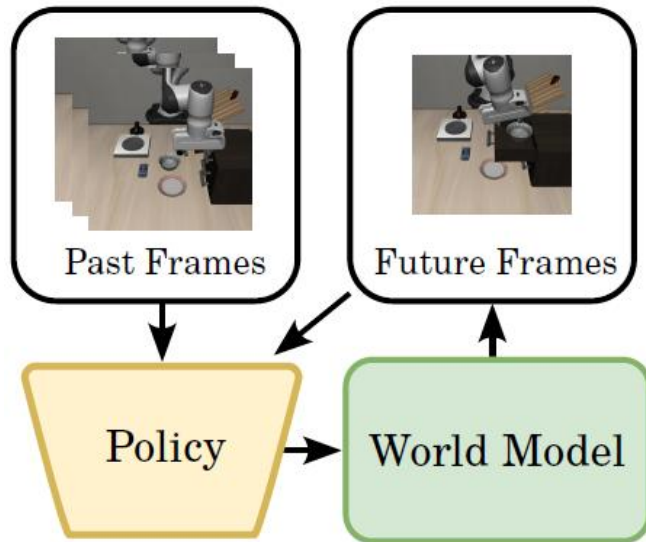
Effective but **~3.7s/step** on RTX A6000 —
— impractical for real-time control



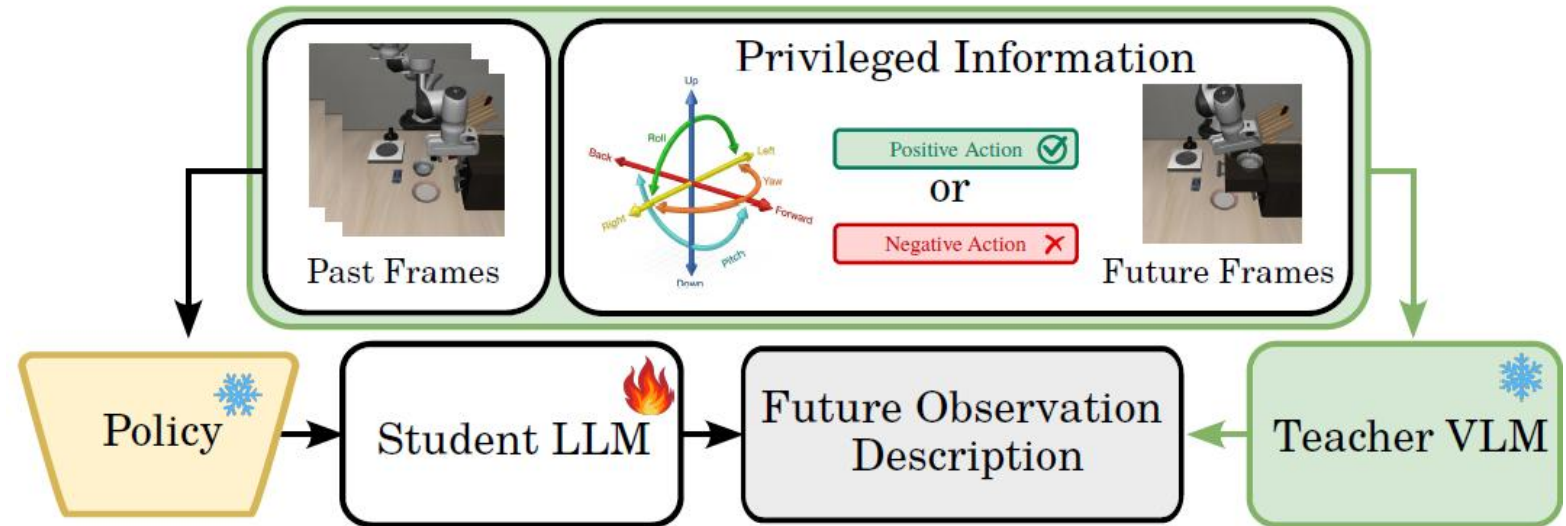
Is visual simulation actually *necessary* for failure prevention?

From *simulate-then-act* to *describe-then-act*

Existing solutions based on world models

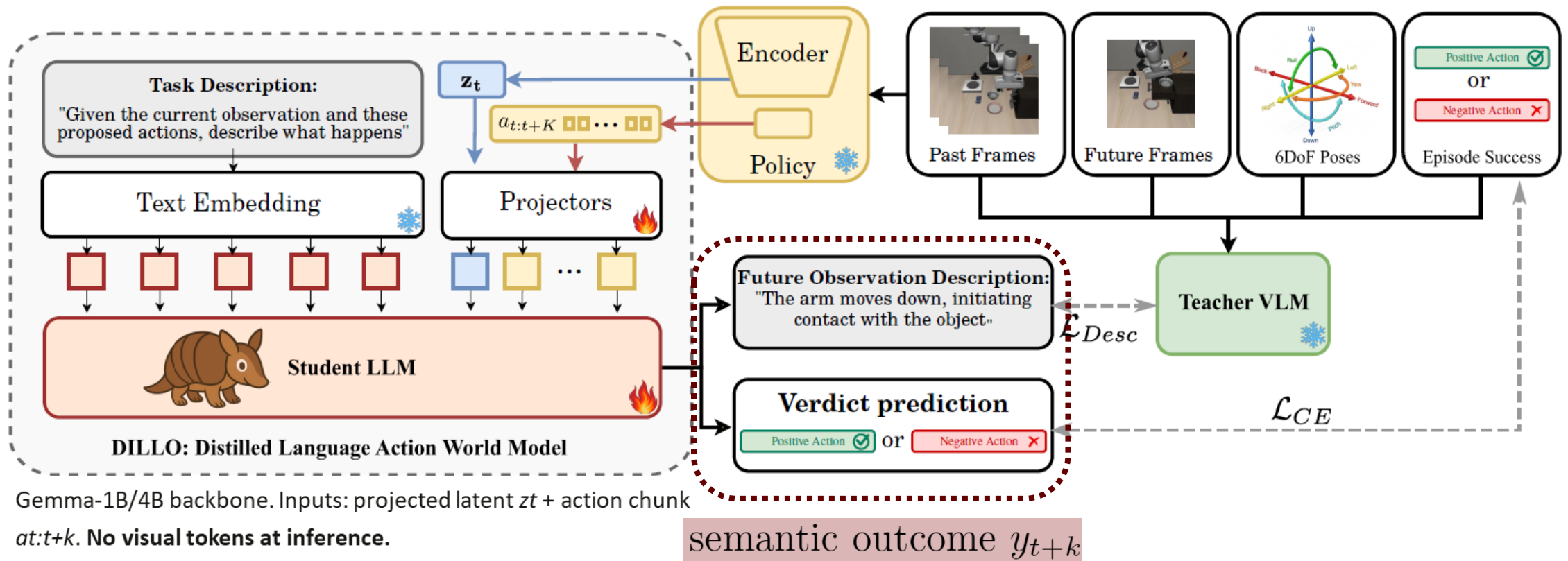


DILLO: **D**isti**LL**ed Language-Acti**O**n World Model



Training DILLO

Privileged access: RGB frames, 6DoF poses, episode outcome.
Annotates offline trajectories with descriptions d and verdicts c .



Steering Success Rates and Verdict Accuracy

Predicted Description d

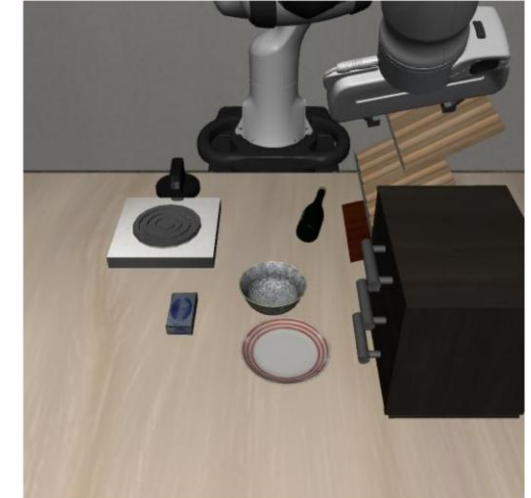
The robot's end - effector moves right, forward, and down toward the plate, aligning the white mug with the target placement area. 


 O_t

 O_{t+K}

Predicted Description d

The gripper moves forward, and up, shifting away from the bowl's position on the table. 


 O_t

 O_{t+K}

Steering Success Rates and Verdict Accuracy

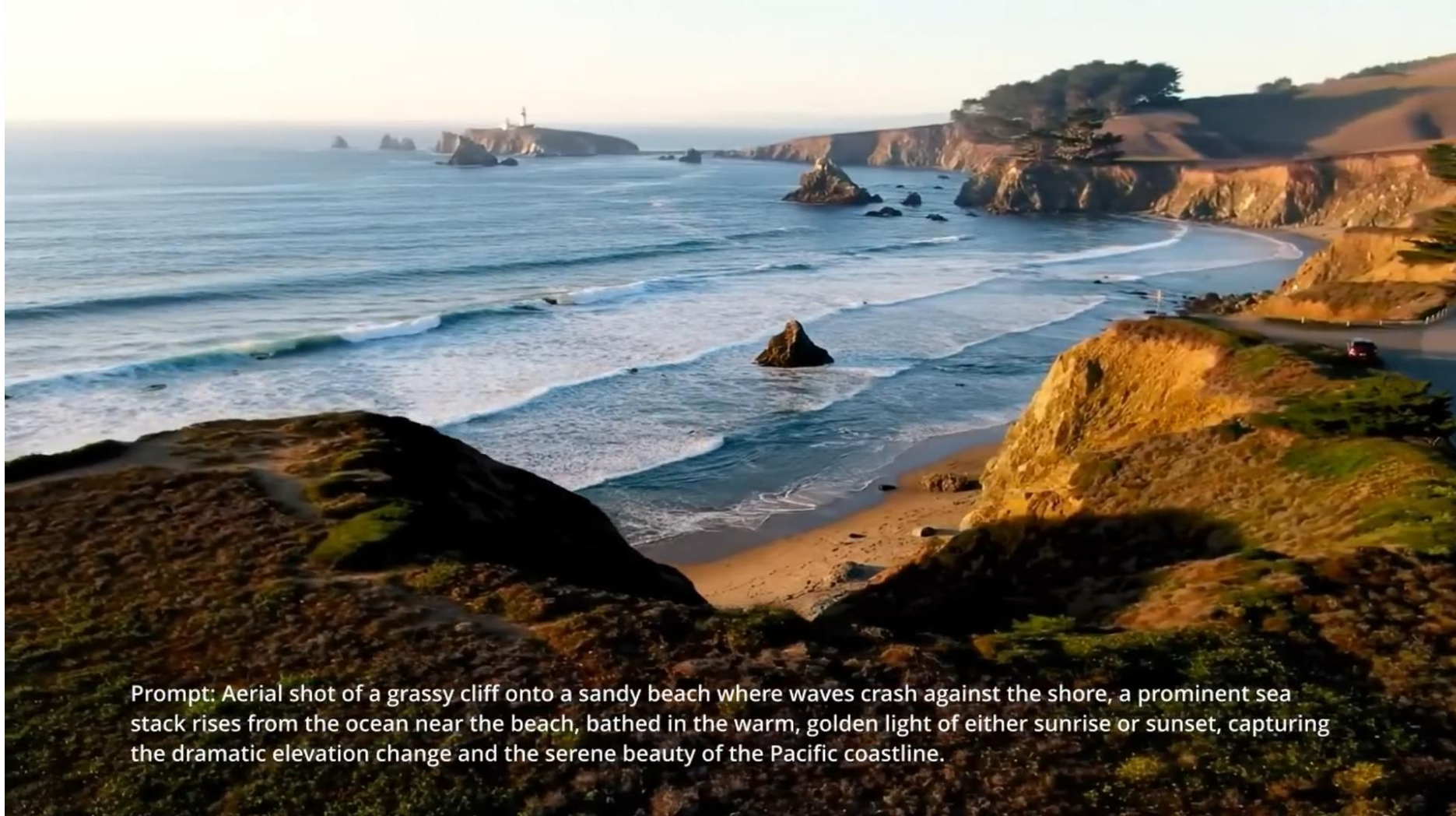
~10x and 14x Speedup

DILLO-1B and DILLO-4B achieve
0.26s/step and 0.38s/step on RTX 3090
vs. 3.7s/step Forewarn* on A6000

		Success Rate			Verdict Accuracy	
	Task	Base	DILLO-1B	DILLO-4B	DILLO-1B	DILLO-4B
MetaWorld	Soccer	65.0%	75.0%	80.0%	82.7%	87.4%
	Sweep-Into	80.0%	90.0%	85.0%	97.7%	92.1%
	Drawer-Open	75.0%	70.0%	90.0%	80.6%	94.7%
LIBERO	Goal	86.0%	87.5%	89.0%	80.0%	87.5%
	Object	82.0%	86.5%	88.0%	79.8%	81.2%
	Spatial	70.0%	76.0%	73.0%	77.8%	86.7%
	10	64.0%	71.5%	74.0%	88.6%	92.5%
	90	71.4%	68.6%	70.5%	86.9%	85.7%

* Wu, Y., Tian, R., Swamy, G., Bajcsy, A.: From foresight to forethought: Vlm-in-the-loop policy steering via latent alignment. In: ICLR wks 2025

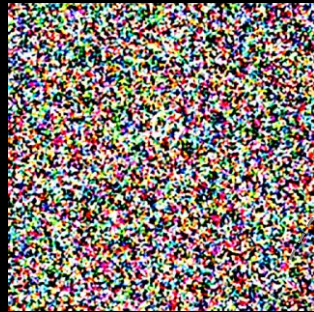
Generative AI



Prompt: Aerial shot of a grassy cliff onto a sandy beach where waves crash against the shore, a prominent sea stack rises from the ocean near the beach, bathed in the warm, golden light of either sunrise or sunset, capturing the dramatic elevation change and the serene beauty of the Pacific coastline.

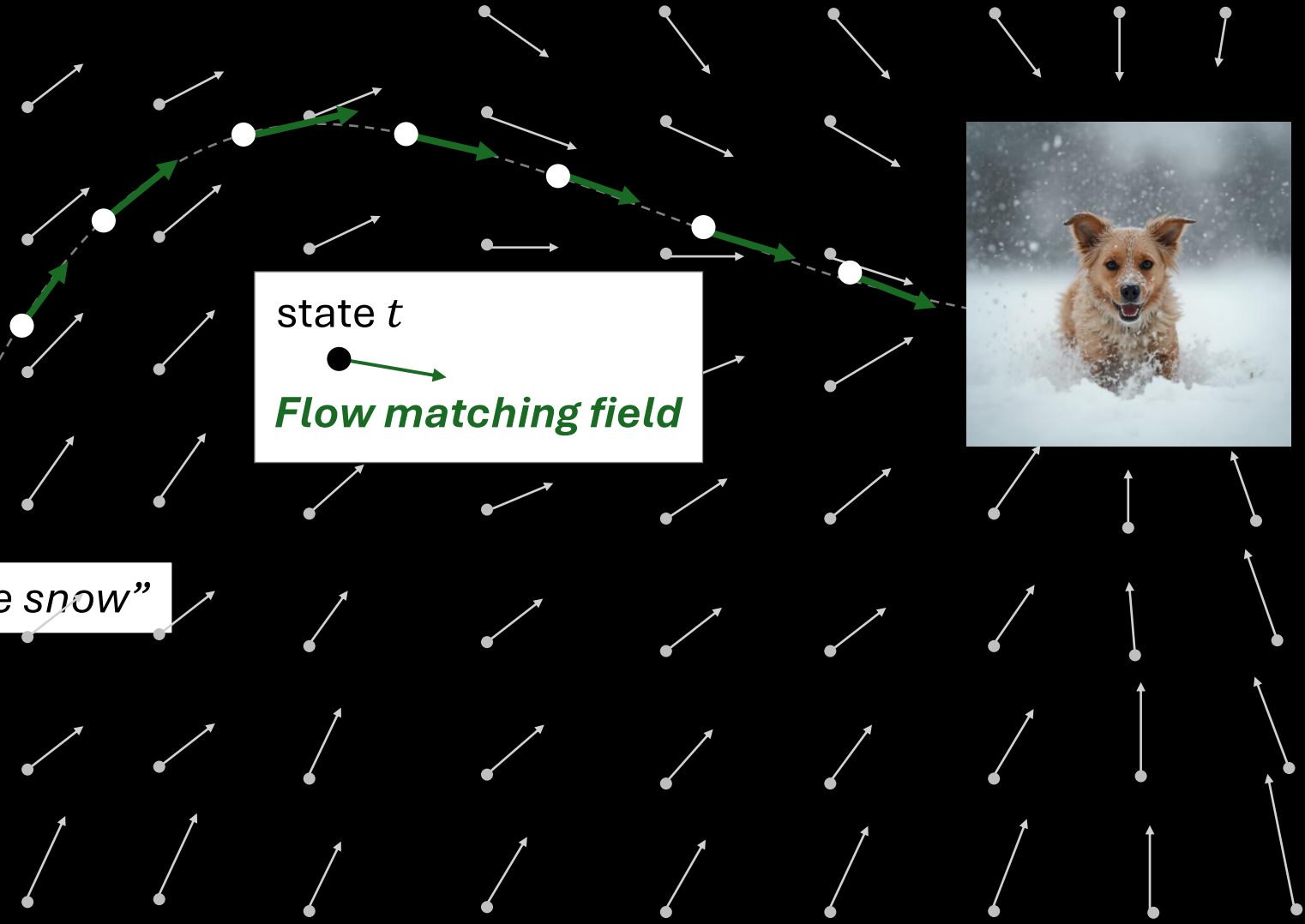
GenAI

FLUX1, SD3.5, ..



+

Prompt: *“a dog playing in the snow”*



GenAI + our editing algebra

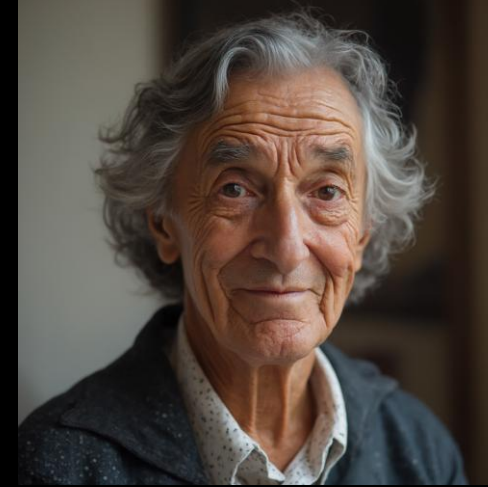
a dog playing in the snow



- *dog* + *cat*



an old person smiling



+ *terminator*



a horseshoe on a anvil



+ *headphones*



a muscular man in gym



+ *alien*



GenAI + our editing algebra

a cat on a couch



+ sunglasses



+ yawning



+ lego bricks



+ striped



a swan in a lake



+ crown



+ dragon



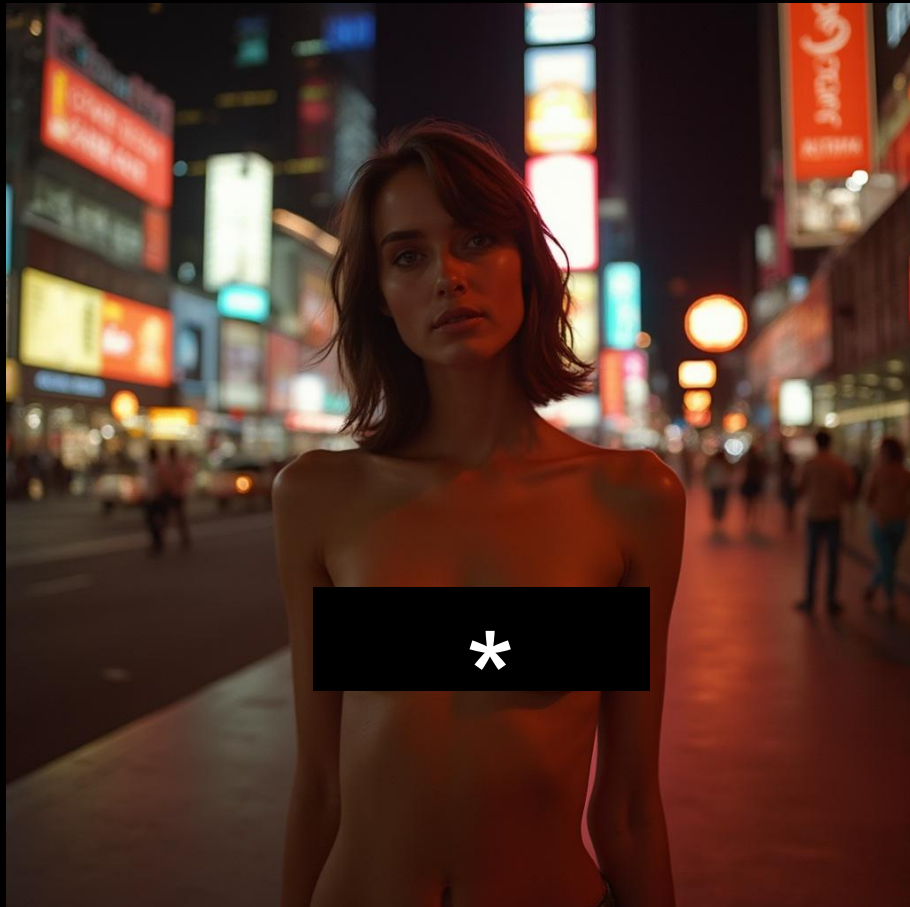
+ origami



GenAI + our editing algebra for safety

In the photograph, the subject is posed nude against a backdrop of city lights, her figure illuminated by the glow of neon signs

- *nude + clothed*



GenAI + our editing algebra for safety

[..input prompt..]

- *nude + clothed*



GenAI + our editing algebra for safety



A toddler climbs an open window ledge on the 20th floor.



Remove "sequential action risk"



GenAI + our editing algebra for safety

"A man drove a motorcycle
blindfolded through
a crowded market"



Remove "sequential action risk"



GenAI + our editing algebra for safety

"A child poured gasoline on the ground and lit a match"



Remove "sequential action risk"



GenAI + our editing algebra for safety

Baseline

Steering Vector
(- nudity)

Ours



Baseline

Steering Vector
(- nudity)

Ours



Human Motion Unlearning

AAAI'26

- Text-to-motion (T2M) has applications in animation, videogame industry, AR/VR etc.
- Increasing interest in T2M for robotics



PyRoki (<https://pyroki-toolkit.github.io>)

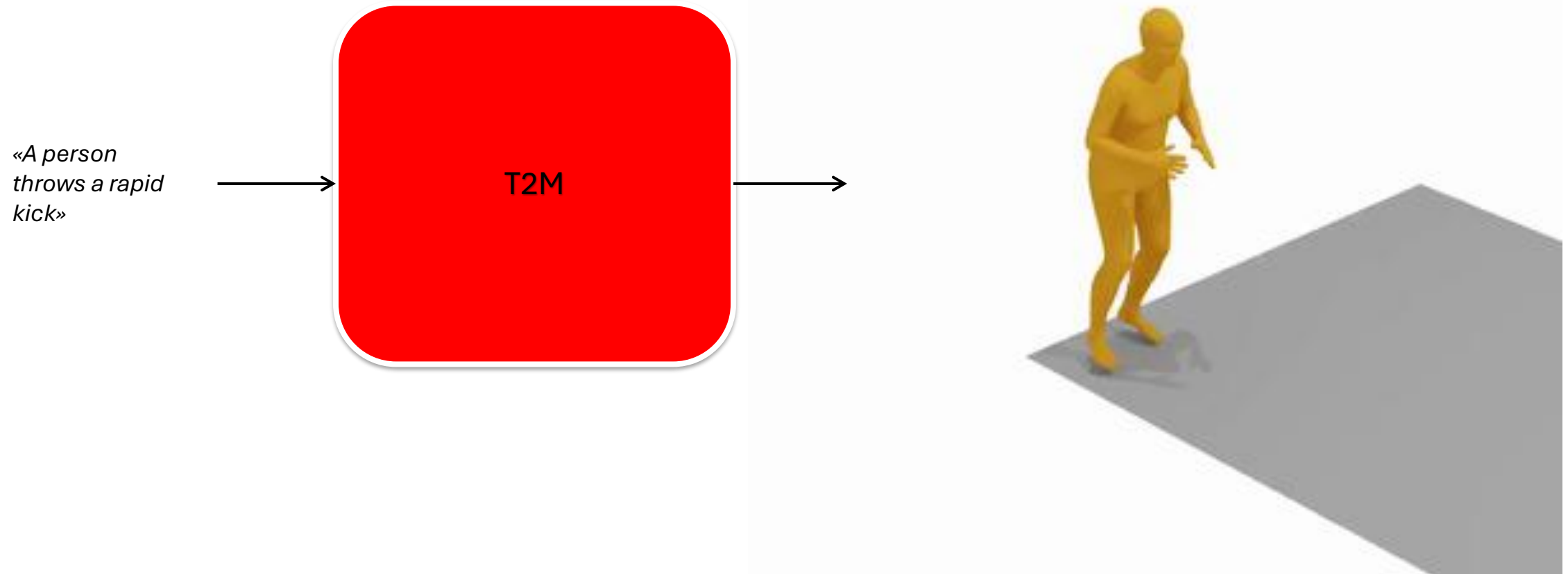


LangWBC (<https://langwbc.github.io>)

Human Motion Unlearning

AAAI'26

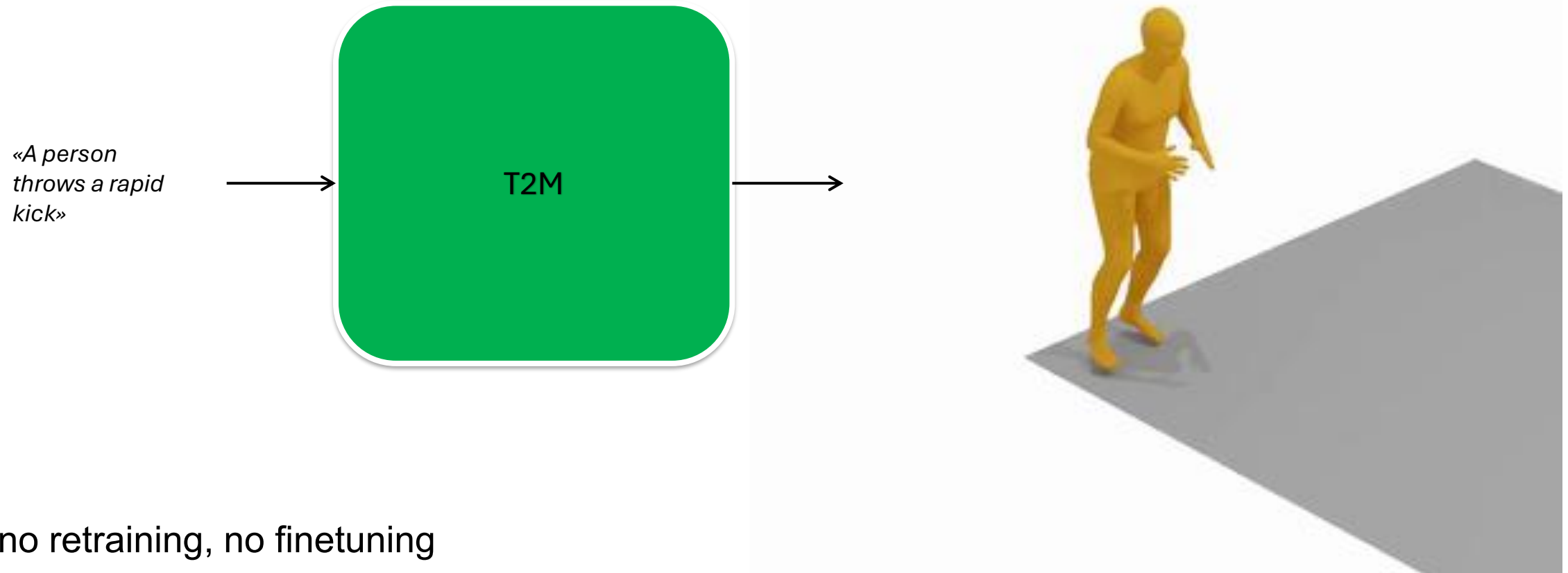
- Consider any human motion synthesis model



Human Motion Unlearning

AAAI'26

- Forget how to generate unsafe motions

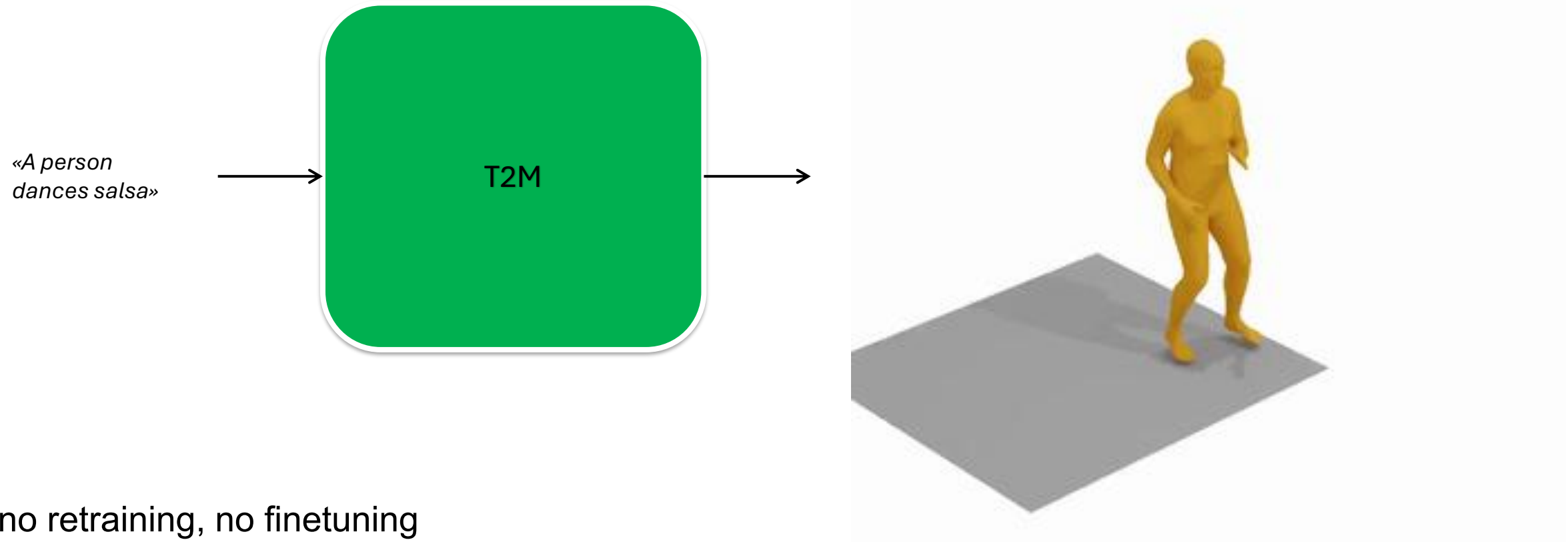


no retraining, no finetuning

Human Motion Unlearning

AAAI'26

- Forget how to generate unsafe motions without affecting safe ones

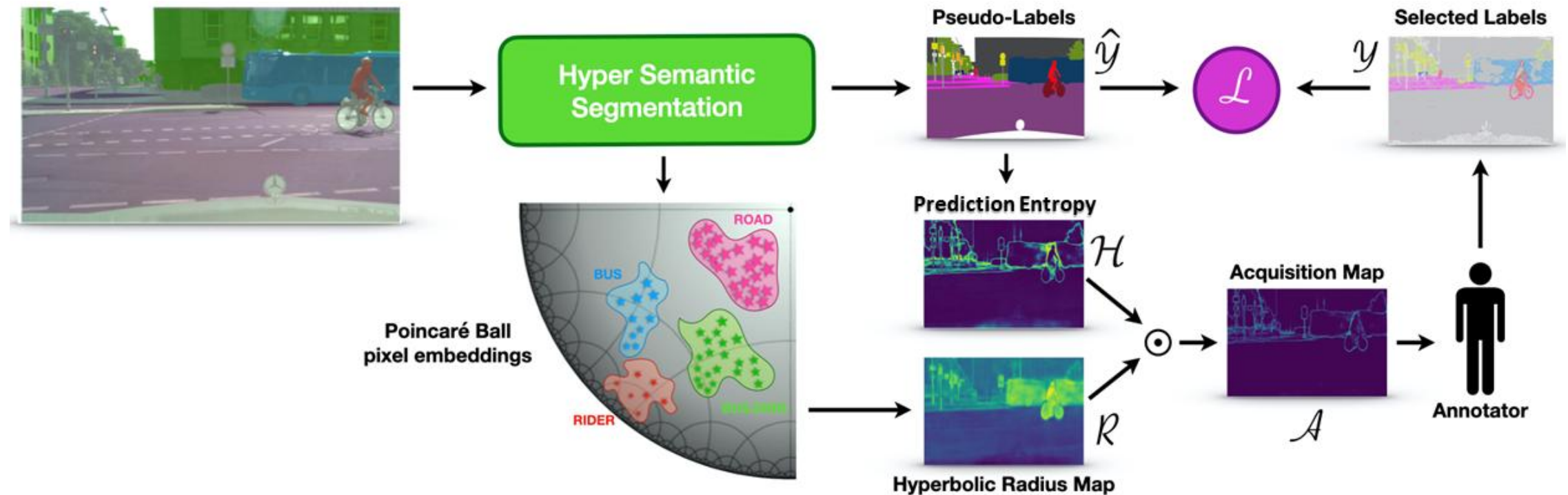


no retraining, no finetuning

Hyperbolic Active Learning for semantic segmentation

ICML'24

- Hyperbolic radius $\approx$ epistemic uncertainty
- Hyperbolic radius: data acquisition strategy



Collaborators:



Funding bodies:



Let's Make AI.



Prof. Fabio Galasso, CTO
Luca Franco, General Manager

ItalAI Story .

Founded in 2024

Sapienza University start-up
Partnership with Panasonic from Day 1

Completing Year 3

15 collaborators, including researchers, scientists, interns, operations and social media managers

Silicon Valley Values

Creating an innovation-focused environment with global vision



Luca Rigazio
Co-Founder - CEO



Prof. Fabio Galasso
Co-Founder - CTO

Luca Franco General Manager PhD, Data Science		Christian Bianchi Lead Researcher MSc stud. AI & Robots		Luca Romani Researcher PhD stud. LLMs	
Matteo Gioia Researcher PhD stud. Robotics		Luca Zhou Researcher PhD stud. CS		Stefano Saravalle Panasonic R&D Intern PhD stud. Robotics	
Matteo Migliarini Researcher PhD stud. CS		Pietro Signorino Panasonic R&D Intern PhD stud. CS		Tommaso Federici Research Intern · MSc stud. CS	
Anastasiia Motyрева Marketing & Communications				Cristiana Polici Operations	

Where our research is published.

Computer Vision & Perception

for product recognition, visual quality control

- **HyCoCLIP** — ICLR'25 (oral, top 2%, w/ Amsterdam Univ.)
- **SEE-ME** ego-pose estimation — WACV'25
- **PREGO** procedural learning — CVPR'24 (w/ Catania Univ.)
- **HALO** hyperbolic MMLM — ICML'24 (w/ Berkeley)

Robotics & Physical AI

for intelligent vending automation, connected machines

- **WIZARD** robot meta-learning — ArXiv'26 (w/ Verona)
- **T-Rex** robotic touch sense — CoRL'26 (w/ Berkeley + Nvidia)
- **LEGO** robot manipulation — ICLR '26 (w/ Berkeley + Nvidia)
- **Hyperbolic social navigation** — IROS'24

Foundation Models & LLMs

for intelligent interaction, dynamic optimisation

- **LongCodeBench** coding LLM — COLM'25 (w/ Stanford)
- **Hyperbolic VLM** — ECCV'24 (w/ Berkeley)
- **SDA** human-robot interaction — ICLR'25
- **SeRpEnt** state space models for LMs — ICML'24

Safe & Trustworthy AI

for cybersecurity, safe deployment

- **Hyperbolic Steering** — ArXiv'26
- **Human motion unlearning** — AAAI'26
- **Video Unlearning** — ICLR'26 (w/ Nvidia)

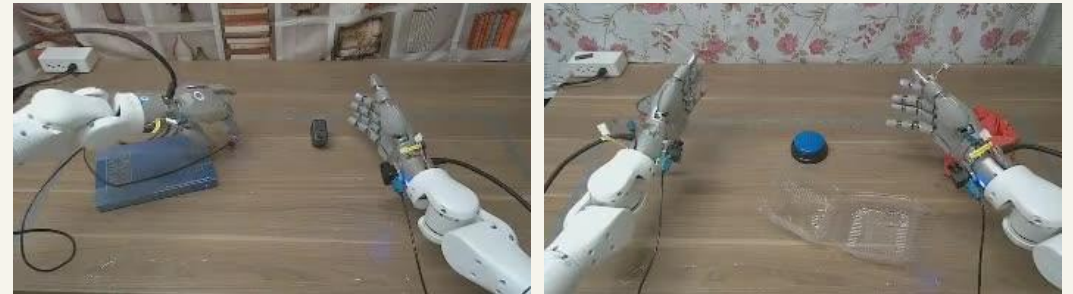
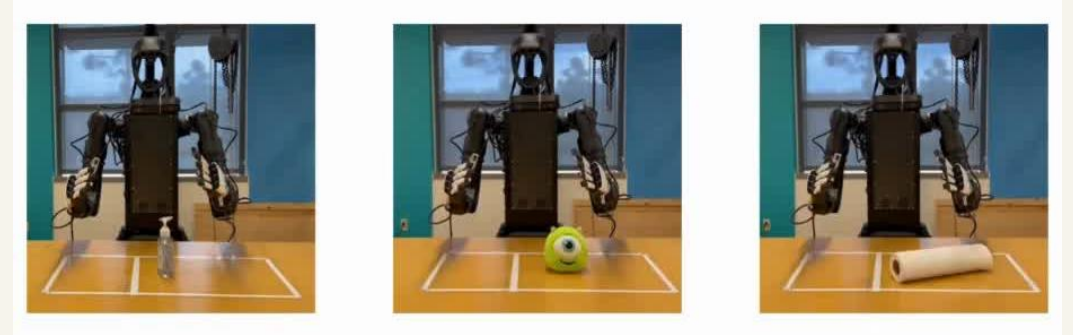
Top-tier venues: CVPR · CoLM · ECCV · ICML · ICLR · IROS · CoRL · WACV · AAAI

ItaAI for Robotics .

Joint research collaboration with the **Berkeley**,
Panasonic, and **NVIDIA**.

Our 2 most recent projects at
ICLR'26 & **CoRL'26**.

More publications to come soon.



Panasonic



ItalAI for LLM .

Research collaboration with Prof. **Hashimoto**,
Stanford University, on LLMs and coding LLMs.

Long-context LLM Coding Agents at **COLM'25**.

More to come on Self-Knowledge Distillation.



ItalAI for Panasonic .

Applied AI research with **Panasonic**.

Solutions for deployment in production environments.

Full pipeline.

Research, prototyping, end-to-end development, deployment, real-world validation.

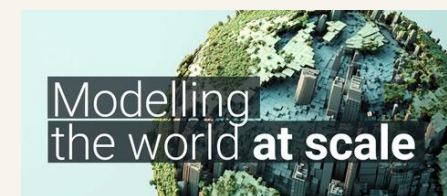
10 students in **3 years** in our **Silicon Valley exchange program**

World-class academic research with real industrial applications.



Open projects for collaborations .

- LLM and Self-knowledge distillation with **Stanford**.
- Robotics with **NVIDIA**, **Panasonic** and **Berkeley**.
- Large-scale, **real-world research** for specific **Panasonic** use cases.
- World Model for **Video Games** with a **Silicon Valley startup which raised \$7M**.



Thanks for your attention



SAPIENZA
UNIVERSITÀ DI ROMA